

Trustworthy Selective Computation for Resource-Constrained Vehicular Edge Networks

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ABSTRACT

Vehicular edge nodes run monitoring models continuously on hardware that cannot compute every frame at full precision. Selective computation—skipping or reusing inference steps—keeps such systems within energy and latency budgets, but it is safe only if the model's uncertainty rises near critical transitions. In continuous driving streams this is not guaranteed: a model trained only for classification can remain confident up to a critical transition, so a confidence-driven scheduler may suppress exactly the frames on which detection depends. We argue that what renders uncertainty usable for scheduling is not calibration but its *temporal organization*—whether the signal separates stable intervals from the moments around risk onset. We study this through the TERA ecosystem: TERA-Gen generates episodes with deterministic, frame-precise onsets; AF-TOI induces temporal organization offline by aligning a compact causal model to an onset-oriented reference; and MHEG exploits the resulting entropy stratification at runtime. In a 2×2 factorial design, adding the scheduler to an unorganized model leaves the miss rate unchanged yet worsens the effective detection delay (TTD_{ef}) by 11.8%, while on AF-TOI-organized trajectories the same scheduler achieves 53.4% cost reduction (TTD_{ef} 0.101 s vs. 0.673 s). A post-hoc calibration control shows that probabilistic calibration and temporally conditioned reliability vary independently. The pipeline emits a discrete operational-risk state ℓ_t at $O(1)$ cost, a candidate input for vehicular edge coordination.

1. Introduction

Consider a monitoring model running on a vehicle's on-board unit. It receives a multimodal sensor stream and must flag a dangerous condition before it escalates, frame after frame, on a low-power microcontroller or automotive-grade accelerator that has no opportunity to offload the computation to remote infrastructure while the event is unfolding [1, 2]. Each frame has to be processed inside a latency budget fixed jointly by the stream's frame rate and the device's energy and thermal envelope; embedded driver-monitoring systems show that real-time inference within such a budget is achievable on automotive-grade hardware [3, 4]. Two consequences follow. First, the operative constraint is not accuracy in isolation but the ability to keep detecting critical events while the node manages its energy and compute over uninterrupted operation. Second, in vehicular and ad hoc edge networks what the node emits should itself be compact and timely

enough to feed downstream coordination. Driver behavior monitoring is a representative instance of this regime, and the one we use throughout. We treat the per-frame budget as given and ask a narrower question: what must be true of the model's uncertainty over time for a scheduler to spend computation selectively within that budget without putting detection at risk? Selective computation, the runtime branch of adaptive computation [5], keeps edge inference inside its resource bounds by conditioning each step on the model's current confidence. SkipRNN [6], AdaFrame [7], ACT [8], and LiteEval [9] report large efficiency gains from this idea, all resting on the same premise: that low uncertainty marks a stable, redundant stretch of input that can be safely skipped. At the edge there is a second requirement – the node's output has to remain useful beyond its own inference loop, since resource-aware monitoring can feed alert dissemination, cooperative perception, and network-level policies – but the premise about confidence is what the scheduler's safety turns on. The premise is usually left implicit: that confidence is a dependable proxy for computational urgency. It is a reasonable one in activity recognition, where a model's uncertainty tends to climb as a new state begins. Continuous vehicular streaming does not guarantee it. A model trained only to separate normal

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from critical, with no reference to *when* risk begins, has no incentive to make its uncertainty behave differently in the seconds before onset than during ordinary driving. Its uncertainty trajectory can stay flat across the transition. A scheduler reading that flat signal then treats an emerging-risk frame like any redundant one and may suppress the recurrent update just as the situation turns, allocating computation away from the critical transition window, at the cost of a missed or late detection. This is the operational risk we are concerned with: not a slightly higher average latency, but the loss of the specific frames on which a critical event is caught. Standard calibration does not detect this failure mode. Probabilistic calibration [10] checks that declared confidence matches empirical accuracy *on average*, pooled over all timesteps; it says nothing about where a given timestep sits within an episode. A model can therefore be well calibrated in this marginal sense and still be unsafe to schedule on, because marginal agreement between confidence and accuracy is silent about the ordering of uncertainty in time – and it is precisely that temporal ordering, near onset, that a causal scheduler consumes. Calibrating the output probabilities cannot supply information the trajectory never encoded. We call the property that a scheduler actually needs *temporally conditioned operational reliability*: the ability of the uncertainty signal to distinguish transition regions from stable ones at the moment the scheduler acts. Neither standard calibration nor existing analyses of non-stationary entropy dynamics [11, 12] treat this as a prerequisite for scheduling. It is also the property that would let a node summarize its situation for higher-level network functions without shipping raw frames or full predictive trajectories. The causal constraint sharpens the problem. On-board inference is strictly causal: a decision at time t may use only the history up to t , so any failure to separate a transition from a preceding stable interval shows up immediately as an unsafe suppression, with no future frame to recover from it. And the episodes are temporally heterogeneous – the same stream mixes gradual, multi-second build-ups with abrupt onsets under 600 ms [1, 13]. On naturalistic data these differences are hard to see at all: an onset annotated to within ≈ 0.31 s at 9.6fps blurs scheduler behavior below that scale, especially when a missed episode carries no explicit penalty [2, 13, 14]. Our hypothesis (Figure 1) is that this cluster of difficulties is addressed less by tuning the scheduler’s thresholds than by deliberately shaping the model’s predictive trajectories during offline training so that they differentiate in time. Once the trajectories carry that structure, the model’s own uncertainty becomes informative about the proximity of a critical event, and the runtime allocator receives an uncertainty signal with operational meaning. On this reading, the apparent conflict between efficiency and coverage is relieved by reorganizing the model’s internal dynamics, not by threshold

engineering – and the same organized signal is what lets the node report a discrete operational state rather than a bare, local confidence value. *By an informative scheduling signal we mean an uncertainty trajectory whose level separates episode phases: low and stable during redundant intervals, rising ahead of and during risk transitions. This is the property depicted in the central block of Figure 1: without it, the gate has no temporal basis on which to discriminate; with it, thresholding entropy becomes a meaningful allocation criterion.* The paper contributes three components that together isolate and exploit this property. *TERA-Gen* (the *TERA episode Generator*, the *synthetic-data component of the ecosystem*) generates synthetic episodes with a deterministic onset calibrated against SHRP2 odds ratios [13]; by fixing the onset instant it removes the annotation variability that would otherwise make frame-precise induction and evaluation impossible. *AF-TOI* is an onset-oriented induction protocol that shapes the causal model’s predictive dynamics so they separate stable regions from transition windows, giving the scheduler an uncertainty signal that means something in time. *MHEG* is the runtime scheduler that consumes that signal; it pairs an unconditional kinematic trigger, which forces computation on abrupt physical change, with an entropic gate whose usefulness depends on the induced organization – and whose behavior when that organization is *absent* is measured directly through a negative control rather than assumed. Beyond choosing between a full recurrent update and state reuse, MHEG emits a discrete operational-risk state $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$ that is low-bandwidth, interpretable, and produced causally. The three components constitute the *TERA (Temporal Evaluation and Risk Assessment)* ecosystem, in which organization is induced offline and allocation happens online on the device. The experiments bear the hypothesis out, and the most telling result is the negative one. On unorganized trajectories, adding the scheduler leaves the miss rate untouched but worsens TTD_{ef} by 11.8%: the scheduler is active, yet without the enabling temporal property it allocates blindly. Give the same scheduler AF-TOI-organized trajectories and it cuts effective cost by 53.4%, with TTD_{ef} of 0.101 s against 0.673 s. A post-hoc calibration control then rules out the obvious alternative explanation: rescaling probabilities improves calibration but does not reorganize entropy in time, so the two properties are empirically distinct. MHEG thus works both as a local selective-computation mechanism and as a source of a compact operational state that could later connect embedded inference to resource-aware coordination. *The remainder of this paper is organized as follows. Section 2 reviews related work and positions our contribution. Section 3 presents the TERA framework: deterministic temporal anchoring (TERA-Gen), temporal organization induction (AF-TOI), and the runtime scheduler*

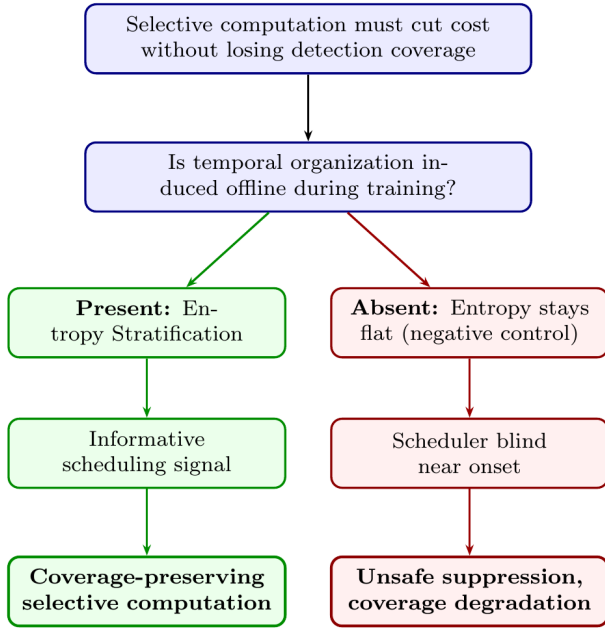


Figure 1: Conceptual hypothesis investigated by TERA. Offline temporal organization induces entropy stratification across episode phases, turning predictive uncertainty into an informative scheduling signal. Without this organization, entropy remains weakly differentiated and selective computation may suppress inference near risk onset. The “informative scheduling signal” block denotes phase-stratified entropy: low in stable intervals and elevated near onset, which is the property the runtime gate thresholds.

(MHEG). Section 4 reports the experimental evaluation, including the factorial study, the robustness controls, and the post-hoc calibration control. Section 5 discusses implications and limitations, Section 6 outlines architectural implications for vehicular edge computing, and Section 7 concludes.

2. Related Work

2.1. Resource-Constrained Edge AI in Vehicular Edge Computing

Vehicular Edge Computing (VEC) and Mobile Edge Computing (MEC) relocate inference from the cloud to on-board or roadside units, trading round-trip latency for tight local budgets in energy, memory, and bandwidth [1, 2, 15]. The motivation is concrete: embedded automotive platforms such as the STM32H7, Cortex-M55, and SA8155P cannot run always-on full-precision inference over a continuous multimodal stream without breaching thermal or energy limits, so some form of runtime allocation is unavoidable. Edge-native ITS work has answered mostly on the model side – quantization [3] and resource-aware architecture [16] – while the broader VEC and MEC literature works at the

network side, on static compression or task placement across nodes. Both leave a complementary question comparatively untouched in safety-critical settings: *when*, within a single continuous on-device stream, is it worth spending a full inference step (Section 2.4)? That single-node question is the one we take up first, because it has to be answered before uncertainty can be relied upon as a coordination signal. A clarification on scope is warranted: vehicular networks (e.g., IEEE 802.11p/DSRC, C-V2X) and general-purpose mobile ad hoc networks differ in protocol stack, mobility model, and infrastructure assumptions, and we do not treat them as interchangeable. TERA operates strictly at the computation layer of a single node – deciding *when* to spend an inference step – and is agnostic to the underlying communication protocol; the operational-risk state ℓ_t it emits is a compact payload that either class of network could disseminate. Protocol-level concerns such as medium access, routing, and dissemination latency are outside the scope of this paper.

2.2. Trustworthy and Validated Edge AI

Efficiency alone is insufficient at the edge; dependability is an equally important requirement. Throughout this paper, “trustworthy” is used in a restricted, operational sense: a selective-computation policy is trustworthy when it preserves critical-episode coverage ($FR \leq 0.05$) while suppressing computation, under the controlled conditions evaluated here. We make no claim regarding broader dimensions of trustworthiness (security, privacy, adversarial robustness), which are out of scope (Section 5). Wang et al. [17] argue that resource limits, heterogeneous environments, and diverse requirements jointly determine whether an edge system can be trusted, and ML-safety work on uncertainty quantification for autonomous vehicles makes the same point from the modeling side [18–20]. The recurring lesson is that dependability has to be demonstrated under deployment conditions, not read off benchmark accuracy. A scheduler that suppresses inference is squarely in this reliability-critical category, since its skip decisions decide whether a safety-relevant event is still caught once computation is cut. What validating such a scheduler requires, concretely, is evidence that the uncertainty signal it reads actually separates transition regions from stable ones – and a direct way to establish the role of this property is to remove it experimentally, which is the function of our negative control. To the best of our knowledge, dependability at this level, the internal structure of the trajectory the scheduler consumes, has not been examined as a validation target in the trustworthy edge AI literature.

2.3. Temporal Risk Detection in Continuous Driver Monitoring

Automated driver behavior monitoring in naturalistic environments is well established. The DMD

dataset [2] provides dense annotations for seven behavior classes with front-camera and physiological fusion, and DADA [14] extends coverage to eye tracking in pre-collision scenarios. Both, however, define onset by retrospective human annotation with variability of ≈ 0.31 s at 9.6fps. That imprecision directly affects edge resource-allocation design. Undetected episodes vanish from latency averages without penalty, and temporal granularity is too coarse to separate transition zones from stable periods. Without a deterministically anchored onset, one cannot tell whether the scheduler invested resources at emergence windows rather than redundant intervals, which is the central decision the edge system must make. The SHRP2 study [13] documents temporally heterogeneous risk trajectories (multi-second gradual transitions down to sub-0.6s abrupt onsets) with up to twelvefold collision risk [1]; dominant validation protocols rely on aggregate recognition metrics and evaluate neither onset timing nor scheduler cost differentiation. A more recent group of systems settles the feasibility question on the hardware we care about. Salakapuri et al. [21] run an integrated distraction and road-object detector on an automotive-grade platform at real-time rates, and lightweight time-series and TinyML models report the same under tight budgets [4, 22]; our own Safe-Drive system pairs behavior detection with seatbelt segmentation for per-frame monitoring [23]. On-device viability has therefore been empirically demonstrated. What these evaluations report, though, is aggregate recognition accuracy and latency – not when a risk is detected relative to its onset, and not what it costs to suppress computation during the transition that leads up to a critical event. The metrics required to expose allocation errors at the scheduler level are therefore not reported. The combination that would expose them – a deterministic onset, an explicit penalty for a missed episode, and suppression treated as a resource-allocation choice – is what TERA-Gen and TTD_{ef} are designed to supply.

2.4. Adaptive Inference and Selective Computation in Embedded Systems

Adaptive computation families such as dynamic networks [5, 24], conditional computation and early exiting [25–28], and split computing [29, 30] cut inference cost by adapting to the input. Two strategies dominate: structural compression (quantization, pruning [3]) and confidence-based scheduling (SkipRNN [6], AdaFrame [7], ACT [8], LiteEval [9], selective classification [31]). Kaya et al. [32] show that overthinking, where computation continues past correct classification, stems from reliance on instantaneous confidence rather than trajectory structure. The present work sits in the on-device branch, where the decision is not *where* to run inference but *when* to invest it within a continuous stream. These efficiency gains are real, but they rest on the assumption that high confidence signals absent risk

and therefore safe suppression. Recent surveys confirm that output confidence and entropy remain the dominant exit or skip criteria [26]. The assumption holds in activity recognition, where predictive trajectories exhibit uncertainty sensitive to state transitions, giving the scheduler a reliable allocation signal. In vehicular monitoring the assumption breaks down. A model may stay confident just before onset not because the situation is stable but because its internal trajectories lack the temporal differentiation that would distinguish transition zones from preceding stable periods. A scheduler designed for activity recognition may then invest in redundant timesteps while suppressing inference at emergence windows, producing an unsafely optimistic efficiency gain. Current design criteria evaluate scheduling through throughput, latency, and deployment metrics without treating temporal detection coverage as an explicit safety constraint. Whether entropy-driven suppression is safe when a missed critical episode carries a real cost is therefore left open, and it is exactly the case AF-TOI and MHEG are built to address.

2.5. Uncertainty Calibration and Operational Reliability

Trustworthy selective computation needs uncertainty that differentiates transition regions, not just uncertainty that matches aggregate frequencies. Predictive uncertainty estimation is established [33–35], and probabilistic calibration, the alignment between declared confidence and empirical accuracy [10], is an accepted automotive-safety criterion [36, 37], with Minderer et al. [38] showing it does not improve monotonically with architecture. Calibration, though, is marginal: a perfectly calibrated model may still schedule computation unsafely if trajectories do not differentiate uncertainty at transitions. Braverman et al. [11] and TRUST [12] confirm the non-stationary character of entropy dynamics without formalizing near-onset behavior as a scheduling signal. A related question, which the reviewed literature does not seem to have addressed directly, is whether improving global calibration also changes how uncertainty is organized in time. There is a structural reason to expect it does not. Post-hoc techniques – temperature scaling [10], Platt scaling [39], isotonic calibration [40] – rescale the output probabilities without touching the weights, and therefore without touching the internal trajectory that a causal scheduler reads. A monotonic rescaling can align aggregate confidence with accuracy while leaving the ordering of entropy through an episode exactly as it was, so entropic dynamics that were not discriminative before remain so afterward. We are not aware of a study in the adaptive causal scheduling literature that tests this divergence experimentally, and we do so here as an explicit control.

2.6. Temporal Structure Transfer and Sequential Modeling

Temporal structure transfer [41–43] and online sequential architectures [44, 45] advance efficiency and accuracy, but none asks whether the transferred structure enables trustworthy runtime allocation. Conventional transfer objectives spread the alignment signal uniformly, pushing compact models to reproduce average predictive behavior rather than uncertainty dynamics that separate stable from transition regions, so entropy stratification for trustworthy selective computation stays unaddressed. Prior works also target segmented offline clips where the full sequence is available; causal streaming invalidates sequence-level normalization and global attention, so organizational gains from those contexts do not carry over automatically. Explicit induction mechanisms are required. *The methodological distinction of AF-TOI relative to these families can be stated explicitly. Temporal knowledge distillation and sequence-level transfer optimize agreement with a teacher's predictions, spreading the alignment signal uniformly over time; their objective is accuracy or compression. AF-TOI instead optimizes the shape of the uncertainty trajectory around a deterministic onset, through asymmetric temporal weighting (Eq. 8) anchored to t_0 : the training target is entropy stratification across episode phases, not fidelity to teacher outputs. Uncertainty-aware inference methods consume uncertainty at runtime but do not reshape its temporal structure during training, and adaptive/selective computation methods assume an informative confidence signal rather than inducing one. To the best of our knowledge, using onset-anchored structural alignment to create the scheduling signal that a selective-computation gate later consumes is the specific contribution of AF-TOI.*

2.7. Gaps and Positioning

Read across these bodies of work, the same gap recurs, and Table 1 localizes it in the **Temp. Org.** and ΔH^* columns. Driver monitoring established the operational stakes but provides neither a deterministic onset nor an explicit miss penalty, so the cost of suppressing computation at a transition window cannot be measured. The adaptive inference literature delivered the efficiency but inherited the premise that high confidence means absent risk – a premise that fails once the internal trajectory is temporally flat. Studies of temporal entropy dynamics [11, 12] confirmed that entropy is non-stationary but stopped short of treating onset dynamics as a scheduling property. Temporal structure transfer built compact models without asking whether the transferred structure was the kind a scheduler could safely allocate on. In the table, the ΔH^* column is $\times$ for every prior entry and $\checkmark$ only for TERA, and **Temp. Org.** is uniformly $\times$ or $\circ$ across prior work. On the evidence we surveyed, we are not aware of prior

work that uses temporal entropy dynamics near a deterministic onset as a scheduling signal under an explicit miss-coverage constraint, or that treats temporal organization as a *necessary* condition along the chain *Entropy Stratification* $\rightarrow$ *Informative Scheduling Signal* $\rightarrow$ *Trustworthy Selective Computation*. These gaps form a dependency chain. Onset dynamics cannot be characterized without deterministic annotation; entropy does not gain discriminative power at transitions without onset-aware training; schedulers cannot allocate safely without informative entropy. TERA treats temporal organization as the structural precondition for this chain: AF-TOI promotes the trajectory structure required for entropy stratification, and MHEG exploits the resulting signal to allocate computation selectively and safely at the edge. The columns of Table 1 are read as follows. **Onset**: deterministic onset annotation for training and evaluation; **TTD**: detection-latency evaluation relative to event start; **Cov.**: critical-episode coverage with an explicit miss penalty; **Causal**: strictly causal online inference; **Temp. Org.**: temporal organization treated as a first-class analytical object (its induction, analysis, validation, and exploitation); ΔH^* : near-onset entropy dynamics used as a scheduling signal under coverage constraints; **Eff.**: efficiency analysis; **Edge**: embedded or vehicular-edge deployment; **Real**: evaluation on naturalistic data or physical hardware. A $\checkmark$ denotes a property that is present, $\circ$ one that is partial or implicit, and $\times$ one that is absent.

On the use of synthetic data in Table 1 The REAL EVAL. column marks this work with $\times$, an explicit methodological choice rather than an oversight. Controlled evaluation requires TERA-Gen, whose deterministic onset annotation removes the ≈ 0.31 s variability of naturalistic datasets such as SHRP2, DMD, and DADA [2, 13, 14]; at that variability, scheduler differences below 0.3 s would be indistinguishable from annotation noise. Calibrating TERA-Gen against SHRP2 odds ratios preserves the statistical structure of real risk trajectories while removing the confounder that would otherwise block frame-precise attribution of scheduling gains to temporal organization. Embedded driver-monitoring systems demonstrate that real-time inference on automotive-grade hardware is feasible under tight per-frame budgets [3, 4], so the operational setting the scheduler targets is hardware-grounded even though the temporal-organization effect is measured in a controlled synthetic environment. Transfer to naturalistic driving data is the primary direction for future validation (Section 5).

Table 1

Comparison between the TERA ecosystem and representative related work. Column definitions are given in the paragraph preceding the table; marks are ✓ (present), ○ (partial or implicit), and ✗ (absent).

Work	Onset	TTD	Cov.	Causal	Temp.	Org.	ΔH^*	Eff.	Edge	Real
<i>Driver monitoring and risk detection</i>										
Ortega et al. (DMD) [2]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Bouhsissin et al. [1]	✗	✗	✗	✗	✗	✗	✗	✗	✗	✓
Salakapuri et al. [21]	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓
Driver state TinyDL [4]	✗	✗	✗	✗	✗	✗	✗	✓	✓	✓
Cognitive distraction [22]	✗	✗	✗	○	✗	✗	✗	✓	✓	✓
<i>Adaptive inference and selective computation</i>										
Hariharan et al. [3]	✗	✗	✗	○	✗	✗	✗	✓	✓	✓
Campos et al. (SkipRNN) [6]	✗	✗	✗	✓	✗	✗	✗	✓	✗	✓
Wu et al. (AdaFrame) [7]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
Graves (ACT) [8]	✗	✗	✗	○	✗	✗	✗	✓	✗	✓
Wu et al. (LiteEval) [9]	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓
<i>Temporal entropy dynamics and uncertainty calibration</i>										
Braverman et al. [11]	✗	✗	✗	✗	○	✗	✗	✗	✗	✓
Dera et al. (TRUST) [12]	✗	✗	✗	○	○	✗	✗	✗	✗	✓
<i>Temporal structure transfer and sequential modeling</i>										
Zhao & Krähenbühl [44]	✗	✗	✗	✓	○	✗	✗	✗	✗	✓
Xu et al. (LSTR) [45]	✗	✗	✗	✓	○	✗	✗	✗	✗	✓
Zheng et al. [43]	✗	✗	✗	✗	○	✗	○ [†]	✗	✗	✓
Bajestani & Yang [42]	✗	✗	✗	✗	○	✗	✗	✗	✗	✓
This work (TERA)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✗

[†] Partial efficiency evaluation only. TERA is evaluated on synthetic data (TERA-Gen) and a reference GPU; naturalistic and on-device validation remain future work (Section 5). Each work is cited in the corresponding subsection of Section 2.

3. From Temporal Organization to Trustworthy Selective Computation

Embedded risk detection processes signals sequentially without access to future observations, and this causal constraint governs TERA. The on-board model operates strictly causally at deployment, while coverage stability comes from the temporal organization induced offline and then exported to the device. The pipeline ingests a continuous stream $\mathbf{x} = (x_1, x_2, \dots)$, where each $x_t \in \mathbb{R}^D$ encodes visual, kinematic, contextual, and acoustic observations. The output y_t depends only on the history $\{x_1, \dots, x_t\}$, which distinguishes the embedded model from bidirectional architectures. The predictive core is a two-layer unidirectional LSTM with per-step cost $\mathcal{O}(4(DH+H^2))$, compatible with automotive-grade edge budgets. **The sensor stream at the input of Figure 2 is the frame-synchronous concatenation of the four on-board modality groups detailed in Section 3.1: behavioral-visual channels from the cabin-facing detector, kinematic channels (speed, longitudinal accelerations, gyroscope), driving-context channels (road type, precipitation, illumination, traffic density), and acoustic event channels, plus two detector-reliability indicators – yielding $x_t \in \mathbb{R}^{31}$ per frame at ≈ 9.6 fps. At each timestep the MHEG gate inspects only the kinematic variation Δx_t and the buffered entropy $H(p_{t-1})$; the full vector reaches the recurrent core only when the gate fires.** The claim that trajectory organization is what lets MHEG schedule safely is tested, not assumed: it is assessed through entropic separation, coverage stability, allocator selectivity, and

TTD_{ef} across the factorial configurations. AF-TOI supplies the organization offline; the negative control removes it, so its role shows up as the difference between the two. Figure 2 summarizes the operational architecture. The framework separates offline induction from online inference. Offline, TERA-Gen, the non-causal reference model, AF-TOI, entropy stratification, and threshold calibration produce an organized causal model and calibrated thresholds. Online, the deployed causal model and MHEG use this artifact to perform selective computation under coverage-preservation constraints.

Operational problem formulation Table 2 lists six design objectives, of which one arbitrates the central conflict: TTD_{ef}. Traditional latency metrics such as TTD_{det} are computed only over detected episodes and ignore complete misses, which can make a system that misses a fraction of events report a low average latency on the rest. A scheduler may thus reduce apparent cost without changing the miss rate, and a coverage-blind metric would not notice. The effective detection delay integrates timeliness and coverage,

$$\text{TTD}_{\text{ef}} = \text{TTD}_{\text{det}} + \text{FR} \times T_{\text{ep}} \quad (1)$$

where TTD_{det} is the mean latency over detected episodes, FR is the missed-episode proportion, and T_{ep} is episode duration. TERA can be read as a trustworthy selective computation mechanism for safety-critical vehicular edge environments, with operational

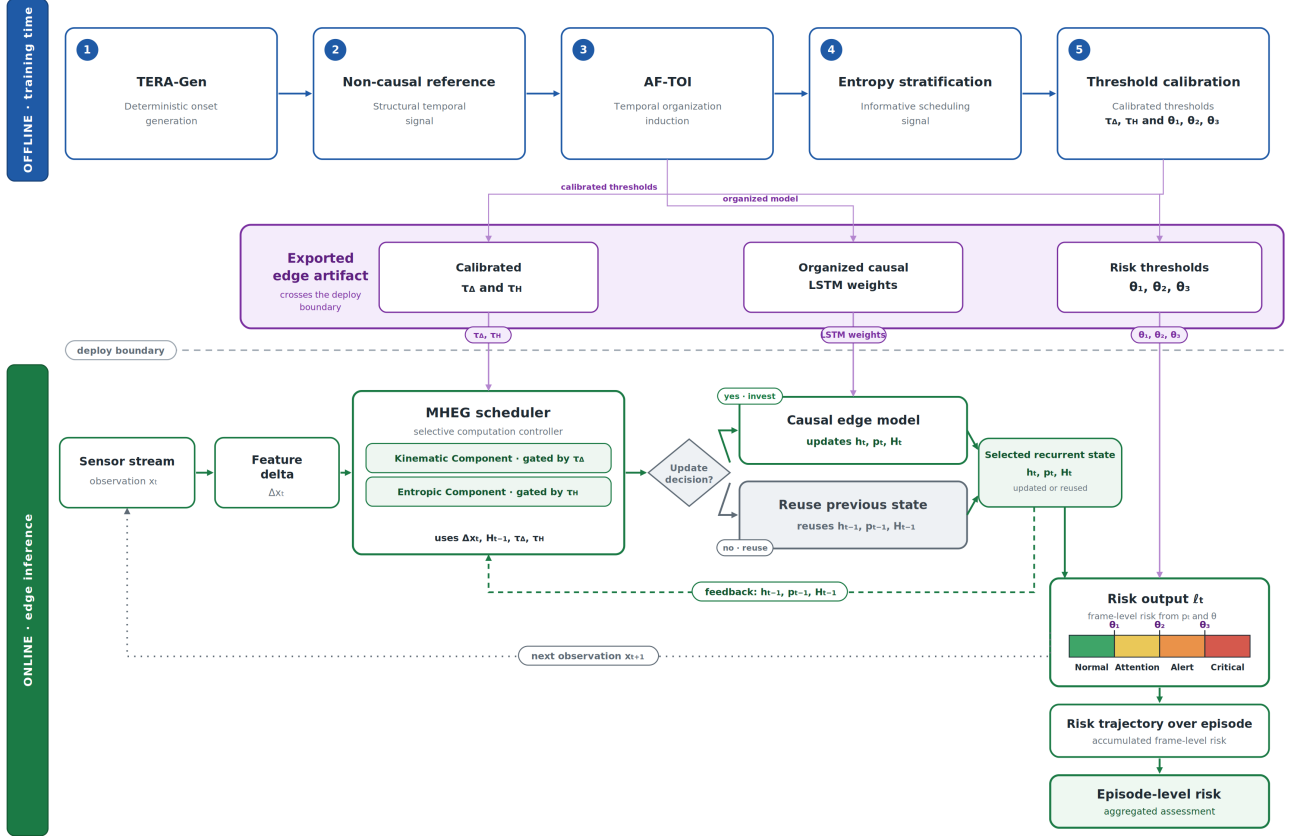


Figure 2: Operational architecture of TERA. Offline stages induce temporally organized trajectories and export the causal LSTM with calibrated thresholds. Online, at each timestep t , MHEG decides whether to update or reuse the recurrent state, from Δx_t , H_{t-1} , and the thresholds τ_Δ, τ_H , under coverage-preservation constraints.

formulation

$$\begin{aligned} \min \quad & \text{TTD}_{\text{ef}} \\ \text{subject to} \quad & \text{FR} \leq 0.05 \end{aligned} \quad (2)$$

Equation 2 abstracts the objectives that guide design and evaluation; it is not an optimization problem solved explicitly. Among configurations satisfying $\text{FR} \leq 0.05$, selective computation is not an independent objective but a consequence of the scheduler identifying redundant regions without compromising critical-event detection. The six objectives are auxiliary detection (Balanced F1 [46]), auxiliary calibration (ECE [10]), auxiliary partial timeliness (gross delay TTD_b [47]), the primary effective timeliness TTD_{ef} (Eq. 1), the safety constraint $\text{FR} \leq 0.05$ [48], and auxiliary computational economy (selective suppression rate, %Sup [6]). Sections 3.1–3.4 follow the causal chain from deterministic onset anchoring through temporal organization induction to trustworthy selective computation.

3.1. Deterministic Temporal Anchoring (TERA-Gen)

Controlled induction and frame-precise evaluation presuppose a deterministic onset as reference. In naturalistic datasets, onset is fixed by retrospective annotation with variability of ≈ 0.31 s at 9.6 fps, which

masks real operational differences between schedulers by making it impossible to tell whether the allocator invested resources at emergence windows. TERA-Gen is a controlled laboratory for this, not a substitute for naturalistic data. By anchoring risk targets to SHRP2 odds ratios it preserves epidemiological coherence while removing the variability that would obscure the effect of entropic differentiation on selectivity. Scenarios span progressive and abrupt risk transitions, enabling frame-precise quantification of TTD_{ef} , miss rate, and robustness. The system models 31 attributes across visual, kinematic, contextual, and acoustic domains grounded in traffic epidemiology.

Input vector structure The vector $x_t \in \mathbb{R}^{31}$ aggregates four domains: behavioral visual (seven driving classes, lateral and gaze deviation), kinematic (speed, longitudinal accelerations, gyroscopic signals), contextual (road type, precipitation, night driving, traffic density), and acoustic (siren, horn, phone ring, background noise). Two further indicators encode instantaneous detector reliability.

Table 2

Unified performance objective matrix of the TERA ecosystem. **Role:** **P** = primary; **S** = safety constraint; **A** = auxiliary evidence. **Meaning** summarizes the physical reading of each metric. [†] Proposed formulation; see Eq. 1.

Objective / Metric	Symbol	Role	Target	Meaning
Multimodal detection (Balanced F1)	F1	A	Max	Harmonic mean of precision and recall over the stream [46].
Predictive calibration (ECE)	ECE	A	Min	Gap between declared confidence and empirical frequency; improves as a byproduct of organization, not as an objective [10].
Partial timeliness (gross delay)	TTD _b	A	Min	Mean latency over detected episodes only; optimistic, as misses are excluded [47].
Effective timeliness	TTD _{ef}	P	Min	TTD _b + FR × T _{ep} (T _{ep} =10.0 s); adds the miss penalty and arbitrates allocation. Proposed [†] .
Coverage stability (Failure Rate)	FR	S	≤ 0.05	Proportion of missed critical episodes; hard constraint on MHEG calibration [48].
Computational economy (suppression)	%Sup	A	Max	Fraction of frames whose LSTM update is suppressed within the FR constraint [6].

Table 3

Risk behaviors calibrated by SHRP2 [13] and target parameterization.

Behavior	Risk Level	OR	risk_target
Normal driving	Normal	1.0	0.050
Hands-free conversation	Attention	1.3	0.074
Mild fatigue / drowsiness	Alert	4.0	0.290
Reaching for moving object	Critical	9.1	0.698
Manual cell phone dialing	Critical	12.2	0.946

Epidemiological calibration Risk targets are projected onto [0.05, 1.00] by

$$risk_target \approx 0.05 + \frac{OR - 1}{12.5} \quad (3)$$

which preserves ordinal severity without claiming to model absolute collision probability. Table 3 gives the resulting values, and the continuous reference label combines mean trajectory and peak intensity,

$$y_{ep} = 0.7 \text{mean}(y_{frame}) + 0.3 p_{95}(y_{frame}) \quad (4)$$

Emergency regimes as design principle TERA-Gen builds critical episodes in two ramp patterns. The *progressive* regime (60% of critical scenarios) simulates gradual development with a latent baseline of 8–20 timesteps and a ramp of ≈ 1.4 s. The *abrupt* regime (the remaining 40%) models fast-ramp perturbations < 0.6 s, the worst case for coverage under adaptive computation, so the scheduler is tested exactly where it must allocate resources to a rapidly emerging risk.

Methodological choice of binary supervision Episodes use binary supervision (Normal vs. Critical). Section 4.7 verifies that this onset-anchored supervision suffices to induce continuous ordinal probabilistic trajectories. Continuous SHRP2-calibrated targets generate temporally structured trajectories; training

labels classify episodes as Normal or Critical, isolating temporal structuring from refined risk regression. Deterministic onset is the precondition for induction, because aligning causal trajectories to a structural signal is only meaningful when the onset instant is fixed and variability-free. AF-TOI operates directly on this anchor.

3.2. Temporal Organization Induction (AF-TOI)

AF-TOI induces temporal organization rather than transferring accuracy. It uses onset-oriented reference-to-target structural alignment during offline training, aiming at predictive dynamics that separate stable regions, pre-onset regions, and transition windows, which is distinct from model compression. Once that differentiation emerges, predictive uncertainty stops behaving as a generic confidence estimate and becomes structured by the temporal phase of the episode. This organization is the enabling property studied here. As trajectories acquire temporal structure, entropy stratifies across episode phases, separating routine intervals from critical transitions, and that stratification is the scheduling signal MHEG depends on. Resource allocation remains a downstream consequence of the induced structure, not a direct output of AF-TOI. The causal constraint of embedded inference motivates the induction: a unidirectional LSTM sees only accumulated history, so abrupt transitions with sub-0.6s pre-onset windows risk weakly differentiated uncertainty and insufficient stratification. To overcome this, AF-TOI uses a non-causal BiLSTM reference with bidirectional access to each episode's full temporal evolution. Free of causal restrictions, the reference develops trajectories that encode the organization of risk transitions across the whole window. AF-TOI transfers this structure to the causal target through offline alignment, so the target reproduces temporally organized dynamics while staying causal at deployment. The induction stays in

the development environment, and only the resulting causal model is exported.

Onset-Oriented Temporal Loss Function The causal target is optimized under the composite loss $\mathcal{L}_{\text{AF-TOI}}$, balancing supervised anchoring with alignment to the reference structural signal,

$$\mathcal{L}_{\text{AF-TOI}} = (1 - \lambda_{\text{TOI}})\mathcal{L}_{\text{hard}} + \lambda_{\text{TOI}}\mathcal{L}_{\text{soft}} \quad (5)$$

where $\lambda_{\text{TOI}} \in [0, 1]$ balances supervision and alignment. The supervised term is binary cross-entropy,

$$\mathcal{L}_{\text{hard}} = -[y_{\text{ep}} \log(\hat{p}_{\text{ep}}) + (1 - y_{\text{ep}}) \log(1 - \hat{p}_{\text{ep}})] \quad (6)$$

Structural alignment is modeled by KL divergence with temperature $T = 2.0$,

$$\mathcal{L}_{\text{soft}}(t) = T^2 \cdot \text{KL}\left(\sigma\left(\frac{z_r(t)}{T}\right) \parallel \sigma\left(\frac{z_s(t)}{T}\right)\right) \quad (7)$$

with $z_r(t)$ the reference logits and $z_s(t)$ the causal-target logits at timestep t . Asymmetric temporal weighting $w(t)$ concentrates induction at onset regions,

$$w(t) = \begin{cases} w_{\text{pre}}, & \max(0, t_0 - k) \leq t < t_0 \\ \exp(-\alpha(t - t_0)), & t \geq t_0 \\ 1, & \text{otherwise} \end{cases} \quad (8)$$

This weighting depends on the TERA-Gen deterministic reference, evidencing the interdependency between ecosystem components.

Three-Phase Structural Alignment Scheduler Direct coupling to the reference signal in early epochs can cause gradient instability, so $\lambda_{\text{TOI}}(e)$ is regulated incrementally,

$$\lambda_{\text{TOI}}(e) = \begin{cases} 0, & 1 \leq e \leq E_{\text{warm}} \\ \frac{e - E_{\text{warm}}}{E_{\text{ramp}} - E_{\text{warm}}}, & E_{\text{warm}} < e \leq E_{\text{ramp}} \\ 1.0, & E_{\text{ramp}} < e \leq E_{\text{total}} \end{cases} \quad (9)$$

with $E_{\text{warm}} = 10$, $E_{\text{ramp}} = 17$, $E_{\text{total}} = 80$ epochs. The scheduler is a **constitutive** part of AF-TOI; removing it compromises coupling stability and the quality of the induced differentiation, as the architectural ablation confirms (Section 4.9). The resulting stratification, with stable intervals at low entropy and pre-onset and transition windows at elevated, varying uncertainty, is the scheduling signal exploited next.

3.3. Trustworthy Selective Computation (MHEG)

MHEG reads the stratification that AF-TOI produces; it does not create it. That dependence is deliberate and consequential: the scheduler's usefulness

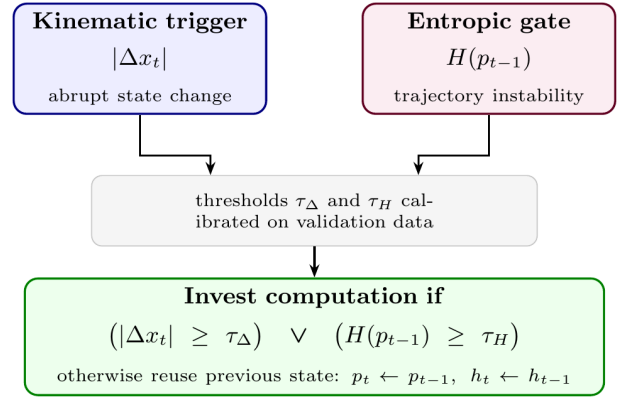


Figure 3: MHEG runtime computation scheduler structure. The kinematic component is an unconditional resource allocation trigger that forces LSTM computation whenever physical state changes abruptly, regardless of entropic state. The entropic component is the trajectory-dependent scheduling signal. When temporal organization is absent, entropy remains undifferentiated and the scheduler loses temporal discrimination power; when induced by AF-TOI, entropy stratifies near onset and the scheduler can allocate resources to emergence windows while safely suppressing redundant intervals.

is inherited from the offline-induced organization, not from the gating rule on its own. When the stratification is absent, the gate has no basis for discrimination, which is precisely the failure reproduced by the negative control. MHEG is the runtime scheduler and adaptive inference controller of the embedded system. At each timestep it decides whether to run a full LSTM update or reuse the previous state, using the AF-TOI-induced organization. Its logic rests on two components, synthesized in Figure 3.

Kinematic Component as Unconditional Resource Allocation Trigger

$$|\Delta x_t| = \max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \quad (10)$$

When this variation exceeds τ_{Δ} , the LSTM core runs regardless of entropic state. The unconditional trigger guarantees allocation in abrupt episodes with sub-0.6 s pre-onset windows, where entropic differentiation may not manifest before the transition begins.

Entropic Component as Trajectory-Dependent Scheduling Signal

$$H(p_{t-1}) = -p_{t-1} \log_2 p_{t-1} - (1 - p_{t-1}) \log_2 (1 - p_{t-1}) \quad (11)$$

When entropy exceeds τ_H , the scheduler invests full LSTM processing. Here entropy is a decision variable reflecting trajectory instability, not an error indicator. The scheduler does not need globally calibrated confidence; it needs uncertainty with temporal differentiation. Absent global calibration does not compromise

allocation, but absent temporal differentiation does. The discriminative power of the entropic component depends on trajectory organization. Without AF-TOI-induced differentiation, entropy sits in a narrow range and the scheduler loses selectivity; with it, phases stratify along the entropic axis and the criterion gains temporal discrimination. This dependency is the causal relationship at the center of the hypothesis, verified in Section 4.

Joint Threshold Calibration With causal-target weights frozen, a per-seed grid search over the validation set selects the threshold combination that maximizes suppression while keeping $FR \leq 0.05$. Calibration uses TERA-Gen synthetic data; online re-calibration is a necessary extension for production.

Scheduling Criterion via Logical Disjunction

$$\max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}| \geq \tau_\Delta \quad \vee \quad H(p_{t-1}) \geq \tau_H \quad (12)$$

When neither condition holds, the LSTM update is suppressed and previous values propagate. The disjunction reflects the asymmetric cost of safety-critical edge systems, where suppressing computation at a transition window is far costlier than a redundant inference step.

Operational Regimes Threshold magnitude determines three regimes. As thresholds $\rightarrow 0$, the system runs full continuous inference: nothing is suppressed, cost is maximal, FR is zero. As thresholds $\rightarrow \infty$, suppression is total and equivalent to halting all LSTM updates: cost is zero, FR is maximal. Between these, per-seed calibration on validation data yields the trustworthy hybrid regime the ecosystem occupies, maximizing economy under $FR \leq 0.05$. The next subsection describes online execution.

3.4. Runtime Inference and Risk Assessment

Algorithm 1 formalizes runtime execution at the edge device. The continuous probability p_t is mapped to four discrete risk levels.

$$\ell_t = \begin{cases} 0 & \text{Normal, } p_t < 0.15 \\ 1 & \text{Attention, } 0.15 \leq p_t < 0.38 \\ 2 & \text{Alert, } 0.38 \leq p_t < 0.50 \\ 3 & \text{Critical, } p_t \geq 0.50 \end{cases} \quad (13)$$

The per-timestep complexity of Algorithm 1 is dominated by line 5. Lines 2–4 and 8–10 cost $\mathcal{O}(D)$ and $\mathcal{O}(1)$, respectively (a maximum over $|\mathcal{K}| \leq D$ channel differences, one closed-form binary entropy, and three threshold comparisons), and involve no matrix operation. The recurrent update of line 5 costs $\mathcal{O}(4(DH+H^2))$ and executes only when the disjunction of line 4 fires, so the expected per-frame cost is $c_{\text{gate}} + (1-s) \mathcal{O}(4(DH+H^2))$, where s is the suppression rate (%Sup) and c_{gate} the

Algorithm 1 Efficient inference cycle of the MHEG runtime computation scheduler

Require: Weights θ_{LSTM} ; thresholds $\tau_\Delta, \tau_H, \theta_1=0.15, \theta_2=0.38, \theta_3=0.50; h_0=0; p_0=0$
Ensure: Level $\ell_t \in \{0, 1, 2, 3\}$ emitted at each timestep t

```

1: for each frame  $x_t$  do
2:    $\Delta x_t \leftarrow \max_{k \in \mathcal{K}} |x_t^{(k)} - x_{t-1}^{(k)}|$ 
3:    $H_t \leftarrow H(p_{t-1})$ 
4:   if  $(\Delta x_t \geq \tau_\Delta)$  or  $(H_t \geq \tau_H)$  then
5:      $(p_t, h_t) \leftarrow \text{LSTM}(x_t, h_{t-1}; \theta_{\text{LSTM}})$  [Invest computation]
6:   else
7:      $p_t \leftarrow p_{t-1}; h_t \leftarrow h_{t-1}$  [Reuse buffer – suppress computation]
8:   end if
9:    $\ell_t \leftarrow \mathbf{1}[p_t \geq \theta_3] \cdot 3 + \mathbf{1}[\theta_2 \leq p_t < \theta_3] \cdot 2 + \mathbf{1}[\theta_1 \leq p_t < \theta_2] \cdot 1$ 
10:  Emit  $\ell_t$ 
11: end for
```

constant gate overhead. The worst case (no suppression) equals the full LSTM cost plus $\mathcal{O}(D)$; the best case is the gate alone.

3.5. Computational Complexity and Edge Deployment

This subsection checks whether the ecosystem is compatible with resource-constrained deployment under strict offline–online separation. Table 4 summarizes the computational profile and the separation between training and device. The highest-cost modules (TERA-Gen, the non-causal BiLSTM reference, AF-TOI, and MHEG calibration) stay offline. At runtime only the causal LSTM target, the MHEG scheduler, the recurrent state, and the calibrated thresholds are active on device. The hidden and cell buffer needs exactly 1,024 bytes of static RAM. The causal LSTM with two layers and 64 hidden units has ≈ 58 K parameters (≈ 228 KB in FP32, ≈ 57 KB in INT8), with expected compatibility with STM32H7, Cortex-M55 with Helium, Jetson Nano, and Qualcomm SA8155P.

Table 4

Computational complexity profile of TERA ecosystem components.

Component	Functional role	Complexity	Executes at	On device
TERA-Gen	Deterministic onset episode generation	$\mathcal{O}(E \cdot T \cdot D)$	Training only	✗
Non-causal BiLSTM reference arch.	Source of temporal structural signal for AF-TOI alignment	$\mathcal{O}(8(DH+H^2))/\text{step}$	Training only	✗
AF-TOI	Temporal organization induction in causal target architecture	$\mathcal{O}(T)$ per episode	Training only	✗
MHEG calibration	Per-seed grid search of scheduler thresholds	$\mathcal{O}(L \cdot E_{\text{val}} \cdot T)$	Training only	✗
Causal LSTM (embedded target arch.)	On-device recurrent inference	$\mathcal{O}(4(DH+H^2))/\text{step}$	Continuous runtime	✓
MHEG (runtime)	Runtime computation scheduler / adaptive inference controller	$\mathcal{O}(1)/\text{step}$	Continuous runtime	✓
Hidden state h_t	Reused recurrent memory (buffer reuse on suppression)	$\mathcal{O}(H)$	Continuous runtime	✓
Thresholds $(\tau_\Delta, \tau_H, \theta)$	Constant scalar scheduling parameters	$\mathcal{O}(1)$	Continuous runtime	✓

Legend: E : training episodes; T : episode temporal length; D : input dimensionality; H : LSTM hidden dimension; L : number of combinations evaluated in grid search.

4. Experimental Evaluation

Validation follows six movements that track the hypothesis, each answering one question and preparing the next. Movements 1 and 2 examine the mechanism: whether AF-TOI observably induces temporal organization in the causal target, and whether that organization confers scheduling selectivity to MHEG. Movement 3 quantifies the operational consequence on coverage, TTD_{ef} , and efficiency. The last three probe the nature of the effect: whether it comes from a reorganized failure distribution, whether it is stable across seeds and regimes, and whether it extends to intermediate risk. Architectural ablation and embedded viability close the section.

4.1. Experimental Protocol

The protocol lets temporal organization be induced, validated, and evaluated with frame-level precision, which is what makes the chain from organization to trustworthy selective computation testable rather than assumed: it renders the entropic differentiation the scheduler depends on directly observable against a deterministic onset.

Synthetic multimodal dataset TERA-Gen synthesizes episodes with 13 training recipes, N01–N03 for normal driving and C01–C10 for critical risk. Six exclusive recipes form the borderline set, A01–A03 for the Attention level and L01–L03 for Alert. Training recipes were generated for the main experimental pipeline; borderline recipes were generated independently and never exposed during training, serving exclusively as a held-out evaluation set for Movement 6 (Section 4.7). Table 5 summarizes the full recipe taxonomy. The dataset consolidates 4,160 episodes across four seeds

Table 5

Recipes used in TERA-Gen synthetic episode generation.

Set	Level	Recipes	Application
Training	Normal	N01–N03	Train, validation, test
	Critical	C01–C10	Train, validation, test
Borderline	Attention	A01–A03	Evaluation only
	Alert	L01–L03	Evaluation only

(42–45) with 70/15/15% split. Each sequence has $T = 96$ timesteps at $\approx 9.6\text{fps}$, $D = 31$, and deterministic onset t_0 at frame-level precision. That onset makes the induced organization verifiable, since it fixes the reference against which entropic differentiation and scheduler selectivity are measured. Table 6 details the full parameterization for reproducibility.

Threshold configuration and calibration Thresholds τ_Δ and τ_H are calibrated per seed via grid search on the validation partition, selecting the configuration that maximizes suppression under $\text{FR} \leq 0.05$. A detection is valid only when $p_t \geq \theta_{\text{TTD}}$ for $m = 3$ consecutive timesteps.

$$\text{TTD}_{\text{det}} = (t_{\text{det}} - t_0) \cdot \Delta t, \quad t_{\text{det}} = \min\{t \geq t_0 \mid p_t, p_{t+1}, p_{t+2} \geq \theta_{\text{TTD}}\} \quad (14)$$

$$\text{TTD}_{\text{ef}} = \begin{cases} \text{TTD}_{\text{det}}, & \text{if } t_{\text{det}} \text{ exists} \\ T_{\text{ep}} = 10.0 \text{ s}, & \text{otherwise (missed episode)} \end{cases} \quad (15)$$

Infrastructure and significance tests Latency is measured in unit-batch mode ($N_{\text{batch}} = 1$) on an NVIDIA

Table 6

Architecture, optimization, and temporal organization induction hyperparameters.

Component	Symbol / Variable	Value	Role in Pipeline
General Architecture and Data	D (input_dim)	31	Multimodal vector x_t dimensionality
	H (hidden_dim)	64	Hidden units per recurrent layer
	num_layers	2	Stacked recurrent layers
	dropout	0.1	Structural regularization factor
Non-Causal BiLSTM Reference Arch. (<i>Offline</i>)	bi	True	Enables non-causal bidirectional processing
	factor	2	Hidden state multiplier ($2 \times H = 128$)
	lr	0.001	Adam optimizer learning rate
	epochs	60	Training epochs with combined $\mathcal{L}_{\text{ref}}$ loss
Loss Function $\mathcal{L}_{\text{AF-TOI}}$ (Causal Target Arch.)	α (onset_exp_alpha)	0.15	Post- t_0 decay rate (Eq. 8)
	T (temp_fixed)	2.0	KL div. smoothing temperature (Eq. 7)
	pre_onset_push_weight	3.0	Amplified weight at latent window
	hidden_align_window	12	Pre-onset window extent (frames)
	λ_{late} (lambda_late)	3.5	Late Penalty multiplier
	θ_{late} (theta_late)	0.15	Causal target reactivity threshold
Structural Alignment Scheduler	E_{warm} (warmup_epochs)	10	Initial epochs with pure hard supervision
	E_{ramp} (rampup_epochs)	17	Linear growth window of $\lambda_{\text{TOI}}(e)$
	E_{total} (epochs)	80	Total causal target optimization epochs
	$[\lambda_{\text{start}}, \lambda_{\text{end}}]$	[0.2, 1.0]	Soft alignment loss coupling bounds

RTX 3050 Ti GPU with PyTorch 2.0 and 100-iteration warmup. Hypotheses use the paired Wilcoxon test [49], with effect quantified by Cliff's delta [50] and 95% confidence intervals by recipe-stratified bootstrap [51].

4.2. Movement 1. Temporal Mechanism

Movement 1 checks, at frame level, whether AF-TOI observably reorganizes the causal target's internal trajectory structure and thereby gives MHEG a temporally discriminative signal. This mechanistic evidence precedes any attribution of allocation gains to organization. In the Baseline, H_t stays poorly differentiated across driving phases: phase distributions collapse into a narrow homogeneous range with no visible separation between stable and transition zones (Figure 4(a)). The scheduling signal is then indiscriminating at the periods of highest criticality, with no temporal basis to distinguish emergence windows from redundant intervals. With AF-TOI, phases stratify along the entropic axis (Figure 4(b)): Normal anchors near zero, Pre-onset converges to the central region ($\frac{\ln 2}{2}$), and Post-onset accumulates at the maximum ($\ln 2$). This is the pattern that gives MHEG a usable scheduling signal. As a byproduct of the induction, alignment between declared confidence and empirical frame-level accuracy improves, with $\text{ECE}_{\text{frame}} = 0.044$ for AF-TOI versus 0.129 for Baseline (Figure 4(c)). This calibration gain accompanies stratification but is not the objective. Figure 5 adds the episode-level view. In AF-TOI+MHEG, suppressed intervals concentrate in low-entropy periods while LSTM computation is preserved near onset, so the scheduler invests at emergence windows and reuses

buffers at redundant timesteps. In the Baseline, suppressions distribute uniformly and the scheduler allocates without temporal basis. **Concretely, each panel of Figure 5 tracks one episode of seed 42 frame by frame: the solid curve is the emitted probability p_t , the dashed curve the entropy H_t that the gate reads, the vertical line the deterministic onset t_0 , the marker ∇ the detection instant, and the shaded spans the frames on which MHEG reused the buffer instead of updating. Reading the AF-TOI+MHEG column left to right: entropy is low and flat through the redundant pre-onset stretch (densely shaded, i.e., suppressed), rises as the transition approaches, forces dense computation around t_0 , and detection follows within a few frames – in both the progressive (top) and abrupt (bottom) regimes. In the Baseline column the same episodes show flat entropy, so suppression lands indiscriminately, including inside the onset window. Under AF-TOI, H_t responds more sensitively to risk transitions, acting as a reliable scheduling variable. Movement 2 tests whether this translates into scheduler selectivity.**

4.3. Movement 2. Scheduler Selectivity

If the reorganization is operationally relevant, MHEG should concentrate investment at the highest-uncertainty windows and reuse buffers at low-entropy timesteps. Figure 5 already shows the resulting asymmetry: with AF-TOI, suppressed intervals concentrate in pre-onset periods while the onset neighborhood is consistently preserved, so computation goes to emergence windows in both regimes. In the Baseline, suppression is uniform and the criterion has no temporal semantics. Figure 6 extends this over 120 critical sequences from seed 42.

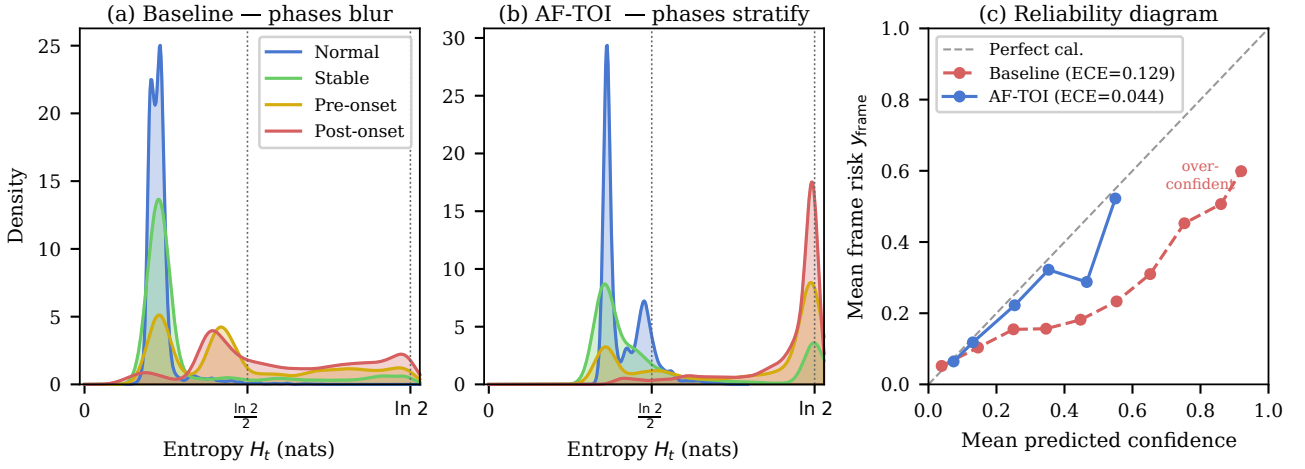


Figure 4: Trajectory organization across driving phases for seed 42. Panels compare Baseline entropy structure, AF-TOI-induced phase stratification, and frame-level calibration alignment ($\text{ECE}_{\text{frame}} = 0.044$ vs. 0.129 for Baseline).

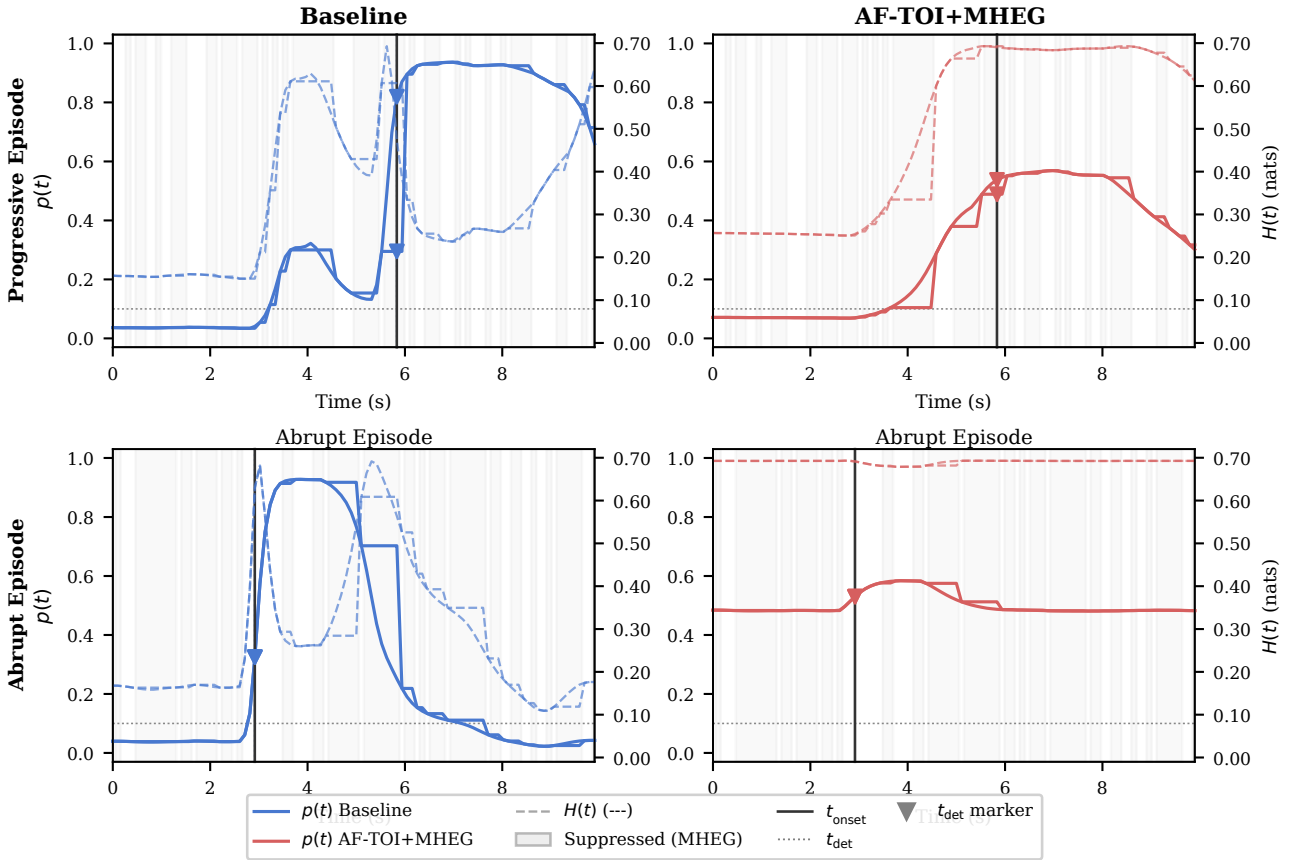


Figure 5: Frame-by-frame temporal evidence for seed 42. Columns show the Baseline and AF-TOI+MHEG configurations; rows show progressive and abrupt episodes. Solid curves denote p_t , dashed curves denote H_t , vertical lines mark t_{onset} , markers ∇ indicate t_{det} , and gray regions indicate MHEG buffer reuse.

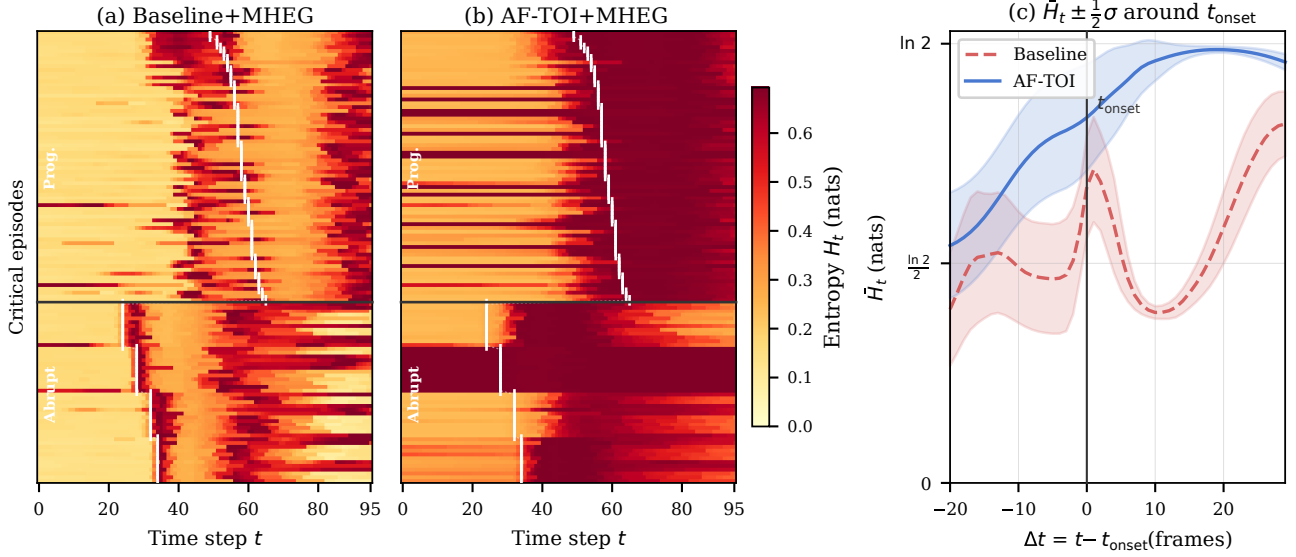


Figure 6: Temporal H_t profile across 120 critical episodes for seed 42. The heatmaps compare Baseline and AF-TOI entropy trajectories around the deterministic onset reference.

In the Baseline, H_t decreases after t_0 , giving no discrimination signal. With AF-TOI, H_t rises near onset, shifting MHEG's investment to emergence timesteps. The alignment between entropic elevation and onset proximity is how the scheduler acquires temporal discrimination. The evidence is consistent with scheduler selectivity tracking trajectory organization: present, it concentrates investment at emergence timesteps; absent, it operates without a temporal criterion.

4.4. Movement 3. Operational Consequence

The 2×2 factorial study isolates the operational consequence on the metrics that arbitrate the central conflict. The negative control is the most informative point, measuring what happens to allocation when selective computation is applied to unorganized trajectories. Table 7 details the design; global results are in Table 8.

The negative control as unsafe adaptive computation Applying MHEG to unorganized trajectories has non-neutral effects on allocation quality. Baseline+MHEG holds $FR = 0.040 \pm 0.052$ at the fixed-baseline level, so it does not reduce misses, and it still degrades TTD_{ef} by 11.8%, from 0.602 s to 0.673 s. The scheduler allocates without temporal discrimination, suppressing emerging-risk timesteps as often as redundant ones. This is the unsafe regime: apparent per-frame cost drops with no coverage guarantee while the effective delay worsens. The result points to the absence of the enabling property, not a scheduler design flaw.

Table 7

2×2 factorial design and per-configuration seed coverage.

Configuration	AF-TOI	MHEG	Seeds	Methodological Role
Baseline Fixed	✗	✗	4/4	Basal reference without temporal organization
Baseline+MHEG	✗	✓	4/4	Negative control: unsafe scheduling without temporal org.
AF-TOI Fixed	✓	✗	4/4	Temporal organization isolated; no scheduler
AF-TOI+MHEG	✓	✓	4/4	Complete ecosystem: trustworthy selective computation

Seeds: number of seeds (42–45) producing valid runs per configuration.

The complete ecosystem and trustworthy selective computation Under AF-TOI organization, MHEG operates on differentiated trajectories. The miss rate is preserved at 0.004 ± 0.008 , with TTD_{ef} rising only 0.015 s relative to AF-TOI fixed. The latency impact of scheduling is about $4.7 \times$ smaller in the complete ecosystem than in the negative control (0.015 s vs. 0.071 s), and absolute TTD_{ef} with active scheduling is 0.101 s versus 0.673 s. In this synthetic scenario, that convergence is consistent with efficiency and coverage coexisting once the scheduler has temporally organized trajectories.

Table 8Global factorial study results under continuous causal streaming (mean $\pm$ inter-seed standard deviation).

Model	MHEG	F1 $\uparrow$	ECE $\downarrow$	FR $\downarrow$	TTD _p (s) $\downarrow$	TTD _b (s) $\downarrow$	TTD _{ef} (s) $\downarrow$	ms/f $\downarrow$
LSTM-Baseline	No	0.357 $\pm$ 0.096	0.690 $\pm$ 0.007	0.040 $\pm$ 0.052	0.476 $\pm$ 0.099	0.206 $\pm$ 0.099	0.602 $\pm$ 0.608	1.015 $\pm$ 0.130
LSTM-Baseline	Yes	0.354 $\pm$ 0.101	0.690 $\pm$ 0.008	0.040 $\pm$ 0.052	0.370 $\pm$ 0.150	—	0.673 $\pm$ 0.682	0.400 $\pm$ 0.148
LSTM-AF-TOI	No	0.751 $\pm$ 0.275	0.497 $\pm$ 0.221	0.004 $\pm$ 0.008	1.419 $\pm$ 0.391	0.044 $\pm$ 0.052	0.086 $\pm$ 0.136	0.920 $\pm$ 0.062
LSTM-AF-TOI	Yes	0.751 $\pm$ 0.275	0.497 $\pm$ 0.221	0.004 $\pm$ 0.008	0.970 $\pm$ 0.266	0.060 $\pm$ 0.071	0.101 $\pm$ 0.155	0.473 $\pm$ 0.079

TTD_p: latency in progressive regime. TTD_b: gross latency over detected episodes (partial, optimistic). TTD_{ef}: effective detection delay with miss penalty (primary arbiter). ECE: auxiliary indicator. %Sup: fraction of frames where MHEG suppressed computation (buffer reuse).

Table 9

Cliff's delta and paired Wilcoxon test: Baseline Fixed vs. AF-TOI Fixed.

Metric	$\delta_{\text{Cliff}}\uparrow$	Magnitude	p-value
ECE	+1.000	Large	0.0625
F1	+1.000	Large	0.0625

$n = 4$. p -floor reflects rank saturation, not absent effect. $\delta_{\text{Cliff}} = +1.000$: dominance across all seeds. Magnitudes per Romano et al. [52].

Regime asymmetry in progressive episodes AF-TOI Fixed shows TTD_p = 1.419 $\pm$ 0.391s in progressive episodes, above the Baseline (0.476 $\pm$ 0.099s): reorganizing dynamics at transition regions can delay classification consolidation in slowly developing episodes. The benefit concentrates in the abrupt regime, where the miss rate falls from 0.099 to 0.010, exactly where misses cost most.

TTD_b vs. TTD_{ef} Gross delay counts only detected episodes and reads optimistically because it hides misses. TTD_{ef} corrects this for edge systems where coverage across all episodes matters more than mean latency on successful detections.

Statistical analysis The AF-TOI F1-Score (0.751 $\pm$ 0.275, CV $\approx$ 37%) shows greater inter-seed variance than Baseline (0.357 $\pm$ 0.096), reflecting sensitivity of the alignment process to initialization. With $n = 4$ seeds the Wilcoxon test saturates ranks (p -floor 0.0625), so analysis relies on $\delta_{\text{Cliff}} = +1.000$ and direct operational means (FR 0.004 vs. 0.040; TTD_{ef} 0.086s vs. 0.602s). Table 9 summarizes. **Given $n = 4$, these statistics are reported as descriptive effect-size characterizations; no claim of conventional statistical significance is made anywhere in the paper.** Table 10 stratifies by operational regime, confirming that the complete ecosystem preserves coverage in both progressive and abrupt profiles. Figure 7 synthesizes the trade-off. Baseline Fixed and Baseline+MHEG occupy identical FR positions, so selective computation without temporal organization gives no coverage benefit;

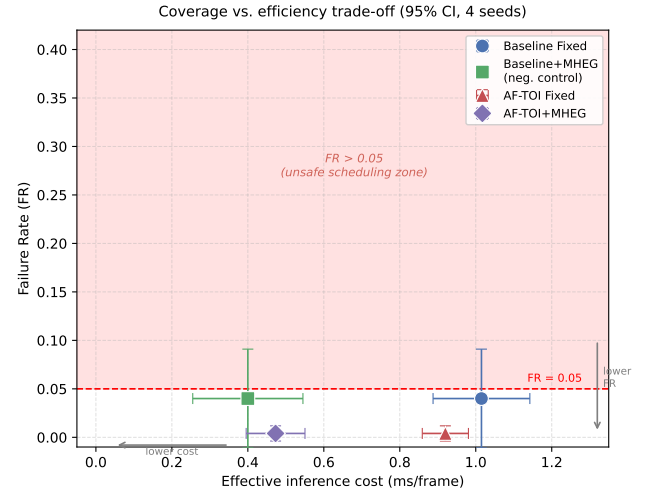


Figure 7: Global miss rate versus effective inference cost per frame (CI 95%, seeds 42–45). The dashed line marks the FR = 0.05 safety boundary; the shaded area denotes the unsafe scheduling region.

AF-TOI+MHEG is the only configuration reaching low cost and FR below the safety threshold at once.

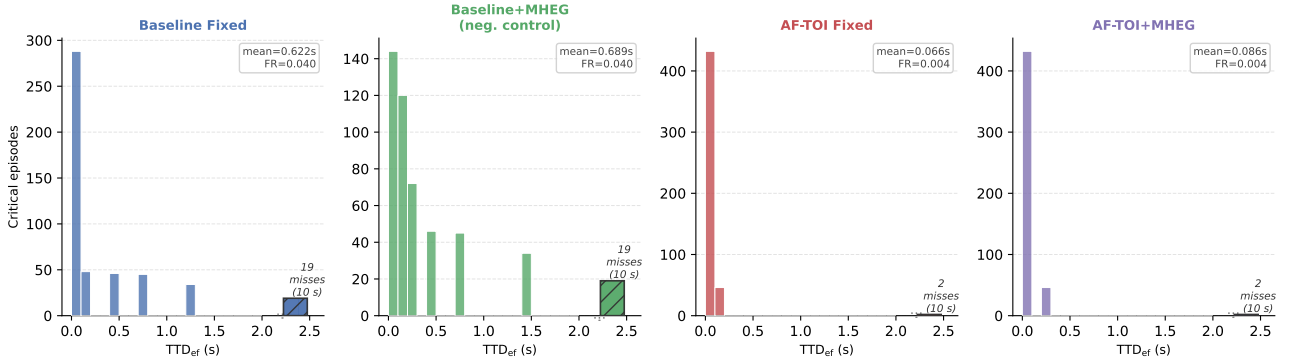
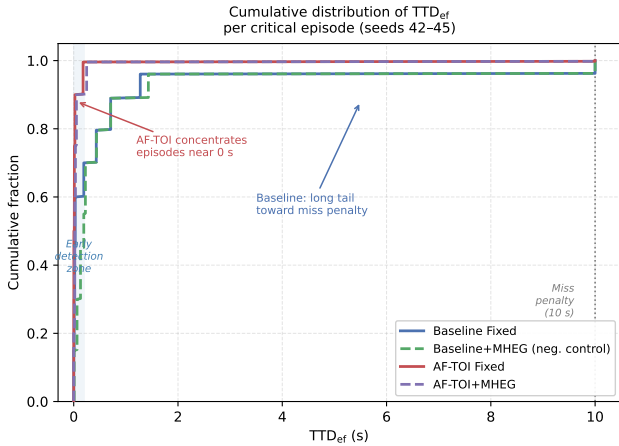
4.5. Movement 4. Distributional Structure

Mean TTD_{ef} can mask heterogeneous per-episode behavior. Figures 8 and 9 show the pattern: AF-TOI histograms concentrate near zero, while variants without organization spread widely, with 19 episodes stuck at the 10s miss penalty, precisely the episodes where the scheduler failed to invest at emergence windows.

The high standard deviations of the non-AF-TOI configurations are not stochastic noise but the 19 severely penalized episodes where computation was suppressed at critical windows. With AF-TOI, the miss population collapses from 19 to 2, so the induction restructures the failure distribution rather than merely lowering mean latency. Baseline Fixed and Baseline+MHEG share identical miss counts, confirming no distributional coverage benefit from selective computation without organization. Adding MHEG to already organized trajectories does not shift the profile, since

Table 10Performance stratified by operational regime (mean $\pm$ sd inter-seed, seeds 42–45).

Configuration	Progressive Regime (60%)		Abrupt Regime (40%)		Global	
	FR	TTD _p (s)	FR	TTD _b (s)	FR	TTD _{ef} (s)
Baseline Fixed	0.000 $\pm$ 0.000	0.476 $\pm$ 0.099	0.099 $\pm$ 0.131	0.544 $\pm$ 0.308	0.040 $\pm$ 0.052	0.602 $\pm$ 0.608
Baseline + MHEG	0.000 $\pm$ 0.000	0.370 $\pm$ 0.150	0.099 $\pm$ 0.131	0.573 $\pm$ 0.352	0.040 $\pm$ 0.052	0.673 $\pm$ 0.682
AF-TOI Fixed	0.000 $\pm$ 0.000	1.419 $\pm$ 0.391	0.010 $\pm$ 0.021	0.046 $\pm$ 0.084	0.004 $\pm$ 0.008	0.086 $\pm$ 0.136
AF-TOI + MHEG	0.000 $\pm$ 0.000	0.970 $\pm$ 0.266	0.010 $\pm$ 0.021	0.063 $\pm$ 0.117	0.004 $\pm$ 0.008	0.101 $\pm$ 0.155

Distribution of TTD_{ef} per critical episode (seeds 42–45)**Figure 8:** TTD_{ef} frequency distributions per critical episode across seeds 42–45 ($n = 480$ per configuration). The 10s value denotes the miss penalty used in TTD_{ef}.**Figure 9:** Cumulative distribution functions of TTD_{ef} per critical episode across seeds 42–45 ($n = 480$ per configuration).

the scheduler suppresses genuinely redundant intervals without creating new failures.

4.6. Movement 5. Robustness

Figure 10 shows performance across four seeds with 95% CI. AF-TOI+MHEG stays below $FR = 0.05$ in every run with lower variability than the other configurations, while variants without organization cross

Table 11Sensitivity to parameter α (AF-TOI without MHEG, seed 42).

α	F1 $\uparrow$	FR $\downarrow$	ECE $\downarrow$	TTD _p (s) $\downarrow$
0.05	≈ 0.88	≈ 0.05	≈ 0.53	≈ 0.40
0.15	0.876	≈ 0	0.324	1.155
0.30	≈ 0.90	≈ 0.008	≈ 0.54	≈ 0.60
0.50	≈ 0.89	≈ 0.010	≈ 0.56	≈ 0.55

the coverage threshold in at least one seed. Trustworthy selective computation thus requires the enabling property to be present consistently. Table 11 shows α sensitivity. The value $\alpha = 0.15$ gives the best balance, with stable F1 while FR and ECE grow more sensitive outside this region, consistent with organization quality depending more on gradient distribution near onset than on global discriminative capacity. Table 12 checks robustness to the miss penalty. The relative ordering across configurations holds for 5, 10, and 15s penalties, so the complete-ecosystem advantage does not depend on the specific T_{ep} .

4.7. Movement 6. Emergent Organization

Movement 6 examines a property not directly optimized: whether AF-TOI organization extends to intermediate risk, producing outputs coherent with SHRP2

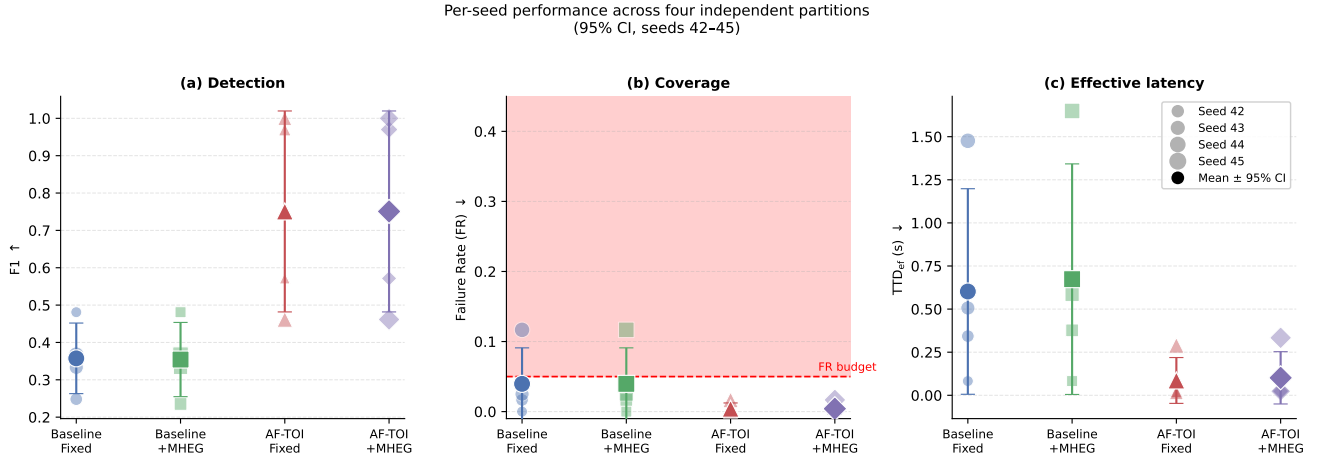


Figure 10: Per-seed performance across four independent partitions (CI 95%). The dashed line marks the $FR = 0.05$ safety boundary.

Table 12

TTD_{ef} sensitivity as a function of miss penalty.

Configuration	Penalty 5.0 s	Penalty 10.0 s	Penalty 15.0 s
Baseline Fixed	0.406	0.606	0.806
Baseline+MHEG	0.478	0.678	0.878
AF-TOI Fixed	0.064	0.084	0.104
AF-TOI+MHEG	0.080	0.100	0.120

Computed analytically from $TTD_{ef} = TTD_b + FR \times T_{ep}$.

Table 13

Probabilistic response in borderline episodes (seed 42, AF-TOI Fixed configuration).

Risk Level	Behavior	$\bar{p}_{max}$	Fraction
Normal	Stable baseline driving	0.006 ± 0.001	0.0%
Attention	Environmental perturbation	0.048 ± 0.195	5.8%
Alert	Mild distraction at low speed	0.380 ± 0.415	48.8%
Critical	Severe distraction with imminent risk	0.271 ± 0.178	70.8%

Fraction: proportion of episodes with $\bar{p}_{max} \geq \theta = 0.10$. Critical episodes from the main test set; Attention and Alert from independently generated borderline recipes.

gradients. In borderline scenarios unseen during training, the Attention regime records $\bar{p}_{max} = 0.048 \pm 0.195$ (5.8% activations above θ) and Alert $\bar{p}_{max} = 0.380 \pm 0.415$ (48.8%), evaluated on seed 42. Table 13 and Figure 11 summarize. The results give preliminary evidence that the induction extends beyond the training recipes. Normal and Attention episodes stay mostly below the activation threshold, while Alert and Critical trigger activation with progressively higher frequency, qualitatively aligned with increasing risk exposure despite no explicit severity objective during training. The Alert–Critical mean ordering is not monotonic (0.380

vs. 0.271), so average probability should not be read as an ordinal severity measure. The large variance in both groups and the heterogeneous recipes suggest that, in this controlled setting, activation frequency was a more stable summary of the system’s response to increasingly hazardous situations. The evidence therefore does not support the claim that AF-TOI learned a severity scale. It suggests instead that the induction produces decision dynamics that grow progressively more likely to activate as criticality increases, even for intermediate-risk conditions unseen in training. Whether this holds on naturalistic data, and whether full ordinal separation can emerge under broader conditions, remain open (Section 5).

4.8. Epistemological Control via Post-hoc Calibration

A complementary question is whether temporal organization could be approximated by post-hoc calibration of Baseline probabilities. Three techniques were applied: temperature scaling (TS) [10], Platt scaling (PS) [39], and isotonic calibration (IC) [40]. MHEG was applied to each variant with thresholds re-calibrated per seed to maximize suppression under $FR \leq 0.05$, yielding Table 14. Probabilistic alignment improves in all variants. Uncalibrated Baseline ECE is ≈ 0.035 ; after isotonic calibration it drops to 0.0075 (reduction 0.0274). The techniques work as intended.

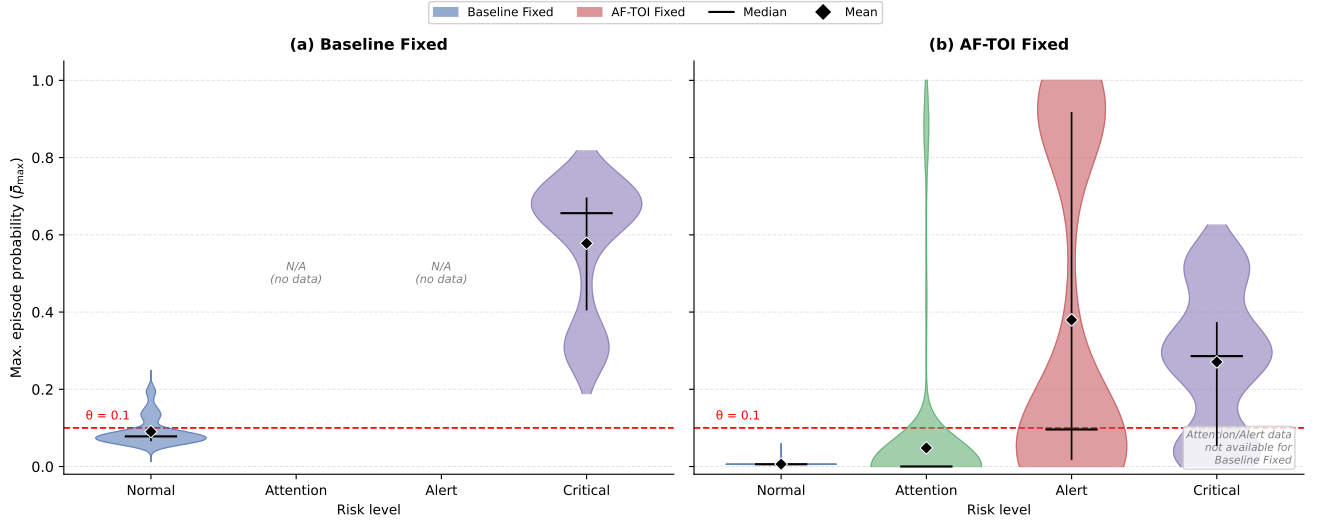


Figure 11: $\bar{p}_{\max}$ distributions by severity level in borderline episodes for seed 42. Baseline Fixed contains Normal and Critical levels; AF-TOI Fixed includes Normal, Attention, Alert, and Critical levels with activation fractions reported in the plot.

Their operational profiles under MHEG differ sharply. Temperature scaling degrades TTD_{ef} from 0.2324 s to 0.4377 s, reproducing the unsafe regime of the negative control. Platt scaling and isotonic calibration reach lower TTD_{ef} under MHEG (0.1221 s and 0.0527 s), which needs a precise mechanistic reading to avoid misinterpretation. The key property is that post-hoc calibration is a monotonic transformation of output probabilities: it preserves the ordering of $p(t)$ along time and cannot reorganize the temporal structure of $H(t)$. Figure 12 confirms this. In panel (a), TS, PS, and IC preserve the Baseline temporal morphology, and the entropy profile around onset stays undifferentiated. The H_{strat} scores match the Baseline across seeds (0.7364 ± 0.1082 for TS, 0.7694 ± 0.0573 for IC), so no technique reorganizes temporal entropy dynamics. In panel (b), AF-TOI structurally shifts the trajectory, elevating $H(t)$ near onset and suppressing it in stable phases. The lower TTD_{ef} for PS and IC under MHEG does not come from temporal reorganization. Both redistribute the probability scale, and IC in particular compresses probabilities toward empirical frequencies, shifting the operating regime of the entropy-based gate without altering the temporal ordering of uncertainty. When MHEG thresholds are re-calibrated to that scale, lower suppression cost is achievable at a given coverage. This is a threshold calibration effect, not an organizational one: H_{strat} stays within the Baseline range and the operational frontier in Figure 13 does not move to a qualitatively distinct location. AF-TOI, by contrast, induces an organization regime not reproduced by probability-scale adjustment, with H_{strat} values that systematically exceed the calibrated Baseline range across seeds. Figure 13 shows the ECE- TTD_{ef} operational locus. Temperature scaling overlaps the Baseline frontier. Platt scaling and isotonic calibration reach

lower TTD_{ef} through threshold recalibration but stay on a frontier structurally distinct from AF-TOI, which reduces FAR and TTD_{ef} simultaneously in a way post-hoc calibration does not reproduce. Figure 14 reinforces this in the pre-onset domain: the pre-onset stratification score stays within the Baseline range across all three techniques and all seeds, while AF-TOI raises it consistently. Probability-scale correction does not modify the internal representation or how it responds to risk proximity; it rescales declared confidence at the output without inducing the temporal differentiation reliable scheduling needs. The distinction matters epistemologically. Reaching low TTD_{ef} through threshold recalibration to a redistributed probability scale differs from reaching it through induced temporal differentiation. In the former, the gain is contingent on the calibrator's particular distribution and may not survive distributional shift, since the gate lacks semantic grounding in the transition dynamics. In the latter, the scheduler exploits a structural trajectory property deliberately induced during training and tied to the causal dynamics of the process. The H_{strat} evidence and the frontier in Figure 13 show the two regimes are empirically distinguishable, with AF-TOI occupying the organizationally grounded one.

4.9. Architectural Ablation of the Non-Causal Reference

To test whether the organization gains depend on the specific BiLSTM topology, an ablation used a simplified protocol (60 epochs, no three-phase scheduler). Two deliberate simplifications explain why the absolute values in Table 15 differ from the main study and must not be compared directly. First, the three-phase alignment scheduler, a constitutive element of AF-TOI that stabilizes coupling and improves induced

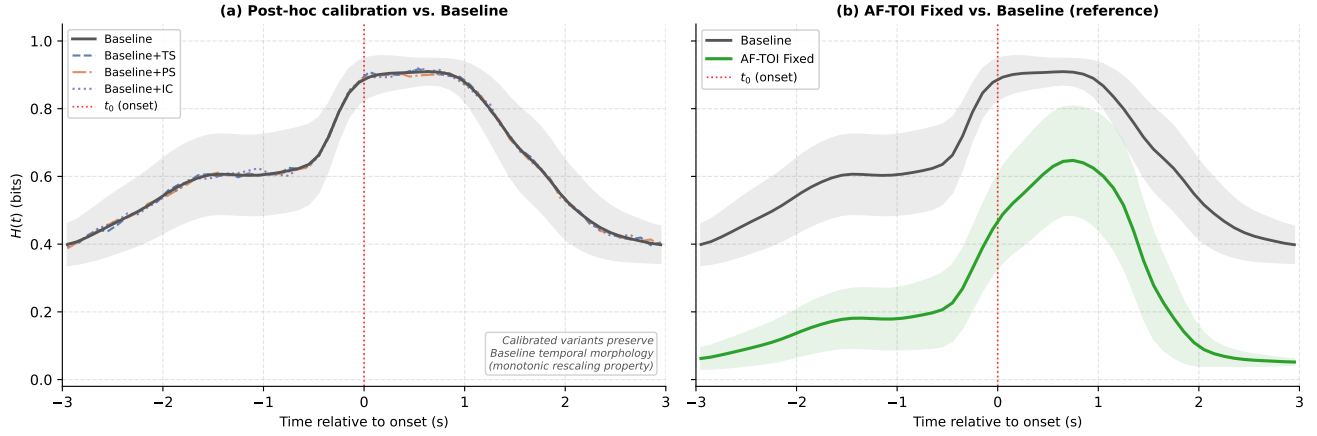


Figure 12: Mean entropy trajectory $H(t)$ around onset for seed 42; aggregate H_{pre} metrics are consistent across seeds 42–45. Panel (a) shows post-hoc calibration variants; panel (b) shows AF-TOI.

Table 14

Epistemological control via post-hoc calibration applied to Baseline with and without MHEG. MHEG thresholds are re-calibrated per seed for each variant. ECE (cal): post-calibration ECE; ΔECE : improvement vs. uncalibrated Baseline (≈ 0.035). Reference: Baseline Fixed FR = 0.040 ± 0.052 , TTD_{ef} = 0.602 ± 0.608 s; AF-TOI+MHEG FR = 0.004 ± 0.008 , TTD_{ef} = 0.101 ± 0.155 s, 48.7% computation suppression.

Configuration	ECE (cal)↓	ΔECE ↑	F1↑	FR↓	TTD_b (s)↓	TTD_{ef} (s)↓	%Sup↑
Baseline+ TS	0.0290	0.0059	0.4410	0.0104	0.1283	0.2324	0.0%
Baseline+ TS+MHEG	0.0326	0.0062	0.4835	0.0250	0.1877	0.4377	58.8450%
Baseline+ PS	0.0184	0.0165	0.4687	0.0000	0.0760	0.0760	0.0%
Baseline+ PS+MHEG	0.0203	0.0170	0.4608	0.0021	0.1013	0.1221	62.0800%
Baseline+ IC	0.0075	0.0274	0.5042	0.0000	0.0362	0.0362	0.0%
Baseline+ IC+MHEG	0.0110	0.0254	0.5040	0.0000	0.0527	0.0527	57.1100%

TTD_b : gross latency over detected episodes (partial). TTD_{ef} : effective detection delay with miss penalty. ΔECE : positive indicates improvement. %Sup: computation suppression rate by MHEG. Lower TTD_{ef} for PS and IC variants reflects threshold recalibration to the redistributed probability scale, not temporal trajectory reorganization; H_{strat} remains within the Baseline range for all three techniques.

organization, was omitted. Second, MHEG was excluded to isolate reference topology, which removes the coverage-preserving layer and drives FR to deployment-unacceptable levels. The question is strictly comparative: which reference topology produces greater relative organization in the causal target under otherwise identical conditions. With the causal LSTM target, TransTOI produces organization comparable to AF-TOI, consistent with induction deriving from the non-causal bidirectional structural signal rather than the specific BiLSTM topology. The causal Transformer target collapses in coverage without external alignment, supporting the causal LSTM as the most robust architecture for embedded deployment.

4.10. Negative Controls and Cost-Matched Robustness

To probe the central claim beyond the 2×2 factorial, we ran a complementary controlled campaign over nine onset/noise scenarios (Table 16), each with

360 episodes across seeds 42–45. The protocol adds two instruments not used in the main study. First, a *compute-rate-matched* budget forces every negative control to spend the same fraction of frames on recurrent updates as MHEG ($\pm 2\%$), so that any difference reflects *where* computation is spent, not *how much*. Second, a near-onset false-suppression metric ($FS@onset$, the fraction of onset-window frames whose update was suppressed) isolates the specific failure that unsafe scheduling produces. Coverage and miss rate are reported as percentages; compute rate is the fraction of frames on which the update runs. These magnitudes are on the stress-protocol scale and are not comparable to the aggregate TTD_{ef} of Section 4.4; the study is designed to rank safety at fixed cost, not to re-estimate efficiency. The budget-matched controls used here (random, unstructured, and lagged allocators) are intentionally harsher than the Baseline+MHEG negative control of Section 4.4: they hold the compute budget fixed and strip temporal structure, isolating the

Table 15

Non-causal reference architectural ablation (simplified protocol, 60 epochs, no 3-phase scheduler). Values not comparable with main study.

Causal Target Arch.	Non-Causal Reference Arch.	Approach	FR↓	TTD _{ef} (s)↓
Causal LSTM	None	Baseline Fixed	0.713 ± 0.069	7.223 ± 0.763
Causal LSTM	BiLSTM arch.	AF-TOI method	0.275 ± 0.189	3.089 ± 1.837
Causal LSTM	Transformer	TransTOI method	0.169 ± 0.163	2.022 ± 1.860
Causal Transformer	None	Baseline Fixed	0.958 ± 0.030	—
Causal Transformer	BiLSTM	AF-TOI method	0.467 ± 0.414	—

Absolute FR values exceed the main factorial study because the scheduler was omitted to isolate the reference architecture effect. TTD_{ef} omitted for Causal Transformer configurations due to near-complete coverage failure (FR ≈ 0.96), rendering the metric uninformative. This reinforces the scheduler's importance for transferred organization quality, without invalidating the non-causal bidirectional structural signal principle.

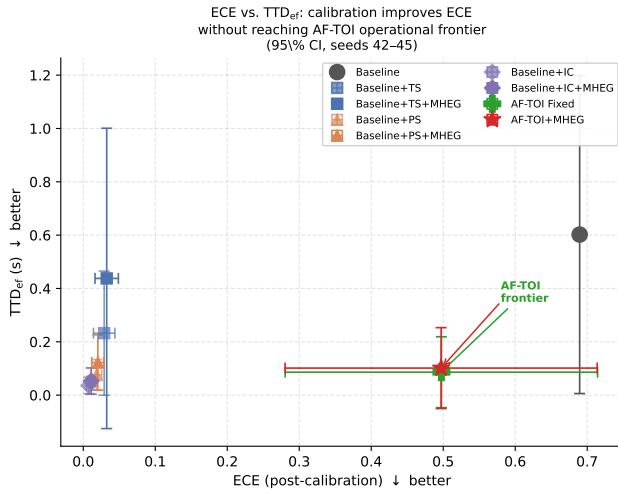


Figure 13: ECE–TTD_{ef} operational locus (CI 95%, seeds 42–45) for Baseline, post-hoc calibration variants, and AF-TOI.

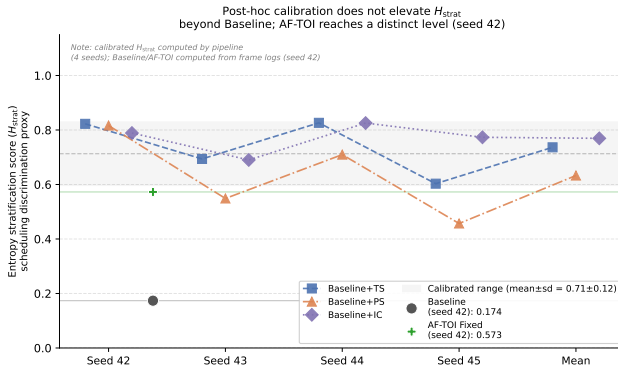


Figure 14: Pre-onset entropy stratification score per seed for seeds 42–45, comparing Baseline, post-hoc calibration variants, and AF-TOI.

effect of *where* computation is placed rather than how much.

Cost-matched safety. The decisive result is that structure, not compute budget, drives safety. Table 17 matches every control to the MHEG budget within

Table 16

Controlled robustness scenarios (each 360 episodes, seeds 42–45).

Scenario	Onset	Entropy	Stressed failure mode
balanced_onsets	mixed	org.	cost saving (reference)
abrupt_onset_dominant	abrupt	org.	false suppression at abrupt onset
progressive_onset_dominant	progressive	org.	delayed detection (TTD _{ef})
rare_abrupt_critical	abrupt	org.	missed transient event (coverage)
high_noise_entropy	mixed	degr.	entropy scheduler misfires
kinematic_signal_missing	abrupt	org.	loss of kinematic protection
entropy_lag	abrupt	degr.	lagged entropy misses onset
false_onset	mixed	org.	wasted compute / false trigger
no_onset_negative_control	mixed	org.	sanity: FS must be ≈ 0

±2%. At *identical* cost, MHEG preserves full coverage with zero onset-window suppression, while budget-matched controls degrade near onset: under abrupt onsets the random and unstructured controls reach 41–44% FS@onset, and under rare transient events coverage collapses to 22–24% (miss rate ≥ 75%) at the same budget. Spending the same computation without temporal structure is therefore unsafe.

Negative-control sweep. Across all nine scenarios, MHEG (hybrid and calibrated) held FS@onset at 0.0 wherever the kinematic channel was available, whereas the budget-matched unstructured (NC2/NC6), random (NC1), and lagged (NC3) controls incurred near-onset suppression on every abrupt-dominated scenario, peaking at 56.7% FS@onset for the lagged control under abrupt onsets. The `no_onset_negative_control` sanity check behaved as required, with false suppression

Table 17

Cost-matched comparison ($\pm 2\%$ compute budget). MHEG vs. budget-matched negative controls. Coverage, miss and FS@onset in percent; compute rate is the fraction of frames updated.

Scenario	Method	Compute rate	Coverage	Miss	FS@onset
balanced_onsets	MHEG (hybrid)	0.217	100.0	0.0	0.0
	NC1 random budget-matched	0.218	99.7	0.3	23.9
	NC2 unstructured aggressive	0.218	100.0	0.0	20.3
	NC6 cost-matched unstructured	0.218	100.0	0.0	20.3
abrupt_onset_dominant	MHEG (hybrid)	0.193	100.0	0.0	0.0
	NC1 random budget-matched	0.195	99.7	0.3	43.9
	NC2 unstructured aggressive	0.196	99.7	0.3	41.6
	NC6 cost-matched unstructured	0.196	99.7	0.3	41.6
rare_abrupt_critical	MHEG (hybrid)	0.066	100.0	0.0	0.0
	NC1 random budget-matched	0.066	21.9	78.1	86.2
	NC2 unstructured aggressive	0.067	24.4	75.6	85.1
	NC6 cost-matched unstructured	0.067	24.4	75.6	85.1

Compute rate is the per-scenario fraction of frames updated under the matched-budget protocol; it is deliberately distinct from, and not comparable to, the aggregate suppression rate (%Sup) of Section 4.4. Within each scenario the budget is matched *across methods* to isolate allocation quality at equal cost.

Table 18

Entropy-lag sensitivity by onset type. Coverage, miss and FS@onset in percent.

Lag	Scenario	Coverage	Miss	FS@onset
1	abrupt_onset_dominant	100.0	0.0	28.3
1	rare_abrupt_critical	100.0	0.0	26.3
3	abrupt_onset_dominant	100.0	0.0	56.7
3	rare_abrupt_critical	64.4	35.6	78.8
5	abrupt_onset_dominant	100.0	0.0	85.0
5	rare_abrupt_critical	0.0	100.0	100.0
10	rare_abrupt_critical	0.0	100.0	100.0

≈ 0 for all methods. When the kinematic channel was removed (`kinematic_signal_missing`), MHEG retained coverage 100% with only 11.5% FS@onset, while the matched controls became unsafe (33–46% FS@onset), confirming the kinematic trigger as the intended fallback for sub-onset windows.

Entropy-lag sensitivity. Table 18 varies the entropic-signal lag. Progressive onsets tolerate lag (coverage stays 100% through 10 frames), but abrupt and especially rare transient onsets degrade sharply: at a 5-frame lag the rare-transient scenario loses all coverage (miss 100%). This is the mechanistic signature of the paper’s thesis, namely that near-onset entropy timing, not aggregate calibration, governs safe suppression.

Ablation of the scheduling channels. A per-channel ablation confirms that both components are necessary. Entropy-only scheduling preserves coverage but adds delay (TTD_{ef} rises relative to the hybrid), while kinematic-only scheduling is catastrophic on progressive onsets, collapsing coverage to 0% because slow transitions never trigger the kinematic gate. The hybrid

matches the always-compute delay at a fraction of the cost, and paired tests over seeds rank the hybrid strictly above the entropy-only control on false suppression (Cliff’s $\delta = -0.5$, large) with no delay penalty against always-compute ($\delta = 0$, negligible).

4.11. Confidence-Threshold and Smoothed-Entropy Baselines

Two additional scheduling baselines complete the comparison requested by the selective-inference literature: a *confidence-threshold gate*, which updates only when the buffered confidence is low (equivalently, when $H(p_{t-1}) \geq \tau$), and a *smoothed-entropy gate*, which thresholds a moving average of the buffered entropy (window $w=5$). Both use no kinematic channel. They are simulated under a replay protocol on the stored full-inference trajectories of the factorial study (seeds 42–45): on update, the buffered probability is refreshed from the stored trajectory; on suppression, it is held. Thresholds are oracle-calibrated on the test partition (maximum suppression subject to $\text{FR} \leq 0.05$, jointly with the detection threshold), an upper bound that *favors* the baselines. Magnitudes are on the replay scale and comparable within Table 19 only.

Table 19 reports two operating points. At the safe point, the purely entropic gates collapse: without an external wake-up signal, a buffered low-entropy value never fires again (the gate deadlocks), so any non-negligible suppression forfeits coverage, and the maximum safe suppression is 0–2% on either trajectory type. Temporal smoothing does not repair this, since averaging a frozen buffer changes nothing. At the matched-suppression point (tuned to spend the same fraction of frames as the entropy+kinematic gate), the confidence and smoothed gates miss 99% of critical

Table 19

Confidence-threshold and smoothed-entropy baselines under the replay protocol (mean $\pm$ sd, seeds 42–45). Safe point: maximum suppression subject to $FR \leq 0.05$. Matched point: thresholds tuned to the suppression level of the entropy+kinematic gate on the same trajectories. Oracle calibration on the test partition favors the baselines.

Trajectories	Gate	Safe point ($FR \leq 0.05$)		Matched-suppression point		
		%Sup $\uparrow$	TTD _{ef} (s) $\downarrow$	%Sup	FR $\downarrow$	TTD _{ef} (s) $\downarrow$
Baseline (unorg.)	Confidence-threshold	0.4 $\pm$ 0.7	0.27 $\pm$ 0.06	97.1 $\pm$ 0.2	0.990 $\pm$ 0.004	10.08 $\pm$ 0.31
	Smoothed entropy ($w=5$)	2.1 $\pm$ 3.6	0.37 $\pm$ 0.21	96.7 $\pm$ 0.5	0.988 $\pm$ 0.005	10.07 $\pm$ 0.32
	Entropy + kinematic	96.1 $\pm$ 0.3	1.90 $\pm$ 0.04	—	0.000 $\pm$ 0.000	—
AF-TOI (org.)	Confidence-threshold	0.0 $\pm$ 0.0	0.24 $\pm$ 0.14	69.7 $\pm$ 1.8	0.594 $\pm$ 0.128	5.95 $\pm$ 1.29
	Smoothed entropy ($w=5$)	0.0 $\pm$ 0.0	0.24 $\pm$ 0.14	69.7 $\pm$ 1.8	0.594 $\pm$ 0.128	5.95 $\pm$ 1.29
	Entropy + kinematic	69.2 $\pm$ 2.1	0.96 $\pm$ 0.40	—	0.000 $\pm$ 0.000	—

The entropy+kinematic row is the reference gate under the same replay protocol; its matched point coincides with its safe point. Confidence and smoothed gates coincide at the matched point because smoothing a frozen buffer does not alter the deadlock.

Table 20

Embedded deployment viability indicators (unit-batch mode).

Indicator	Baseline	AF-TOI	AF-TOI+MHEG
Parameters		≈ 58 K	
FP32 / INT8		≈ 228 KB / ≈ 57 KB	
Recurrent buffer		1024 bytes	
FLOPs/timestep		≈ 28 KFLOPs	
Latency (ms/f)	1.015	0.920	0.473
Computation suppression	0%	0%	48.7%

episodes on unorganized trajectories and 59% on organized ones. The entropy+kinematic gate, in contrast, sustains $FR = 0$ at 96% and 69% suppression, and the benefit of temporal organization appears in its effective delay: 0.96s on AF-TOI trajectories against 1.90s on unorganized ones at comparable safety. This supports two claims at once: the unconditional kinematic trigger is a structural necessity for any buffered entropy gate (Section 3.3), and organization, not the gating rule, is what converts suppression into safe suppression.

4.12. Embedded Deployment Viability

At runtime, the active infrastructure is limited to the causal LSTM target, the MHEG scheduler, constant thresholds, and the recurrent buffer (Table 20). AF-TOI+MHEG reaches 48.7% computation suppression (%Sup) and 53.4% effective cost reduction (ms/f, per-frame latency relative to Baseline Fixed), two metrics that jointly characterize the operational economy. Estimates for dedicated embedded platforms come from scaling projections; physical chip validation is required before production. All embedded-viability figures in this subsection are analytical (parameter counts, memory footprints, FLOP counts) or measured on a reference GPU; none derives from execution on physical automotive hardware.

Experimental evaluation synthesis Taken together, the six movements converge on a single conclusion: in this embedded causal setting, temporal differentiation is what enables coverage-preserving selective inference, with driver monitoring as the concrete case. AF-TOI induces the differentiation that makes MHEG’s criterion discriminative, so computation goes to emergence windows and is withheld through redundant ones; the gain traces to a reorganized failure distribution rather than a lower mean, it holds across seeds, and it carries over to intermediate-risk episodes never seen in training. All of this runs within ≈ 58 K parameters and 1,024 bytes of static RAM – a footprint that automotive-grade microcontrollers and low-power accelerators can carry.

Artifact availability Source code and TERA-Gen synthetic datasets will be made available as described in the Data Availability statement.

5. Discussion

Temporal organization as the enabling condition for edge resource allocation The most informative result is the negative control, rather than the full configuration. Baseline+MHEG holds the miss rate but buys no efficiency in return, which locates AF-TOI’s primary effect precisely: it is not that AF-TOI raises classification scores, but that it creates the conditions under which selective computation is safe at all. Without differentiated dynamics, efficiency and coverage do not merely fail to improve together – they compete, because the scheduler trims apparent per-frame cost while the effective delay climbs. Once the trajectories are organized, that competition relaxes (Movements 4.2 and 4.3): H_t turns into a variable the scheduler can discriminate on, concentrating computation at high-instability timesteps and reusing buffers through the stable ones. The premise behind SkipRNN [6] and ACT [8] – that low uncertainty marks content safe to skip – holds in activity recognition and breaks

in causal edge inference, and the negative control is where that break becomes measurable. In summary, the effectiveness of the runtime allocator depends on a property the model must already possess: temporal differentiation in its trajectories.

AF-TOI as enabler of trustworthy selective computation A model with 95% classification accuracy and an unsafe scheduler is not equivalent to one whose scheduler allocates correctly at emergence windows. TERA shows that scheduler quality is at least as important as classification quality, and that protocols optimizing only accuracy or calibration do not guarantee the trajectory structure scheduling reliability depends on, a direct implication for edge-native ITS design.

Probabilistic calibration and temporal organization as independent edge design properties The post-hoc calibration control carries a direct design implication. Temperature, Platt, and isotonic scaling improve aggregate alignment without touching weights or internal trajectories, and the $H(t)$ morphology around onset stays within the Baseline range for all three, confirming that monotonic rescaling does not induce temporal differentiation. Where PS and IC reach lower TTD_{ef} under MHEG, it is threshold recalibration to the redistributed scale, not organizational restructuring, as the invariant H_{strat} profile shows. The frontier reachable through temporal induction stays qualitatively distinct from what scale adjustment can reach. Marginal calibration and temporally conditioned operational reliability vary independently and have distinct mechanistic origins.

Effective coverage versus gross delay The Baseline shows lower gross delay on detected episodes, but that metric omits misses. For safety-critical edge systems, coverage across all episodes matters more than mean latency on successes, and any metric excluding misses reads optimistically.

Complementary role of the kinematic component When pre-onset windows fall below 0.6s, temporal differentiation may not appear in entropy fast enough. MHEG's kinematic component covers this by triggering allocation unconditionally when instantaneous physical variation signals a sudden state change. The disjunction between components handles two emergency regimes with distinct dynamics without depending exclusively on entropy, which matters for robustness under abrupt onset.

Implications for edge AI deployment and vehicular edge integration Most deployment properties follow from the offline/online separation. The computationally intensive stages – TERA-Gen, the BiLSTM reference, AF-TOI, and threshold calibration – remain in the development environment; the device runs only the

causal LSTM, the calibrated thresholds, and the recurrent buffer, with no reference model to execute and no on-line retraining. MHEG's own overhead is $O(1)$ per timestep, negligible beside the recurrent update it gates. At $\approx 58K$ parameters, a 1,024-byte buffer, and ≈ 28 KFLOPs per timestep, the deployed model sits within the envelope of current automotive-grade microcontrollers and accelerators, and because the only artifacts that move to the device are calibrated weights and thresholds, it is compatible with OTA updates and needs no connectivity at inference time. The ℓ_t signal, produced at $O(1)$ cost, is the natural point of contact with higher-level VEC and MEC coordination – but that contact is an implication we develop in Section 6, not something evaluated here.

Future directions Three extensions follow. First, the MHEG scheduling signal could feed future VEC and MEC offloading policies (Section 6), triggering remote inference when local entropy exceeds a coordination threshold while suppressing redundant timesteps and reallocating the saved energy. Second, online adaptation of MHEG thresholds under distributional shift is the principal step toward production. Third, adversarial robustness under sensor faults, malicious perturbations, and modality degradation is a natural safety extension.

Scope and applicability The campaign is scoped to a single embedded node running continuous causal inference, which isolates the temporal-organization effect from network-level confounders. No networked coordination is evaluated. The mechanism could be reused in future distributed deployments; Section 6 discusses how the MHEG entropy gate could inform VEC and MEC offloading, with empirical validation under real network conditions left as future work. Robustness against adversarial attacks, sensor faults, and cybersecurity threats is out of scope and noted in Section 5.

Methodological rationale for synthetic evaluation The systematic use of TERA-Gen is deliberate. Controlled evaluation of temporal organization needs three properties naturalistic datasets cannot provide together at the required granularity: (i) *deterministic onset annotation*, removing the ≈ 0.31 s variability that would make scheduler differences indistinguishable at 9.6fps; (ii) *causal isolation*, preventing correlated naturalistic confounders from masking the effect of trajectory structure on selectivity; and (iii) *frame-precise attribution*, enabling exact quantification of when the scheduler allocates relative to onset. TERA-Gen anchors synthetic trajectories to SHRP2 odds ratios [13], preserving statistical realism while achieving annotation precision naturalistic collection cannot guarantee. It does not replace naturalistic evaluation; it makes the hypothesis testable at the required granularity. Generalization to

naturalistic data is the primary direction for future validation (Limitation 1).

Threats to validity and deployment limitations

Four limitations bound the claims. First, evaluation relies on TERA-Gen synthetic data, designed to isolate the causal relationship between temporal organization and trustworthy selective computation under controlled onset; confirmation on naturalistic datasets such as DMD [2] and DADA [14] remains necessary for external validity. Second, MHEG thresholds are calibrated on synthetic data and may need re-calibration under real-world shift from driver variability, illumination, and sensor noise. Third, the factorial analysis uses four seeds (42–45); with $n = 4$, the Wilcoxon signed-rank test has a minimum one-sided p -value of 0.0625, which prevents conventional significance at the 0.05 threshold, so the Wilcoxon and Cliff’s delta estimates are read as descriptive effect-size characterizations rather than definitive inferential tests. The strongest evidence is therefore the directionality and mechanistic consistency of the negative control, where MHEG on unorganized trajectories worsens TTD_{ef} while the same scheduler on organized trajectories preserves coverage and cuts computation. Expanding to 10–12 seeds with an a priori power analysis is a priority for the next iteration. Fourth, deployment feasibility is assessed through profiling and analytical estimation rather than execution on physical automotive hardware, so latency, energy, thermal behavior, and sensor-noise robustness remain to be characterized on STM32H7-, Cortex-M55-, and SA8155P-class platforms. **A related caveat is that a controlled environment could, in principle, favor an onset-oriented training strategy. Three design elements mitigate, without eliminating, this risk: episodes mix progressive and abrupt regimes in fixed proportion (Section 3.1); Movement 6 evaluates recipe families never seen in training; and the stress campaign of Section 4.10 spans nine onset/noise scenarios, including degraded entropy, lagged entropy, missing kinematic channels, and false onsets. Generalization across risk-trajectory families beyond these, and under real distribution shift, remains to be established on naturalistic data.**

Planned real-data validation As the primary next step we will test whether the temporal organization induced on TERA-Gen transfers to real driving video, using the DADA2000 dataset [14], which provides frame-level accident annotations – an abnormal-onset frame, an accident frame, and an abnormal-end frame – together with annotated contextual attributes (weather, lighting, scene, and road geometry). The abnormal-onset frame serves as a real, frame-precise onset t_0 , and the interval up to the accident frame is the emergence window in which detection must occur; this is the property naturalistic driver datasets usually lack and that TERA-Gen was built to supply synthetically.

The scene and contextual attributes map onto the visual and contextual channels of our feature schema. Because DADA2000 is scene-facing and carries no ego-vehicle kinematics, the kinematic trigger of MHEG has no native input; we will therefore evaluate the entropic gate both on its own and with an optical-flow-based kinematic proxy, an ablation that isolates the contribution of temporal organization from that of the kinematic trigger. Evaluation will use clip-level splits stratified by scene and accident type, with clips from the same source kept on the same side to prevent leakage, and MHEG thresholds re-calibrated per split on real validation data and never transferred across domains. We will compare three transfer regimes – zero-shot, with the organization-bearing recurrent layers frozen and only the input projection adapted; light fine-tuning, with the recurrent layers frozen and the input and output layers adapted; and full re-induction of AF-TOI on the real training split – against the same negative control used here (Baseline+MHEG). Success is defined not by matching the synthetic magnitudes, which are on a different scale, but by reproducing the qualitative relationships established on TERA-Gen: the negative control failing to improve coverage, and AF-TOI+MHEG preserving coverage while suppressing computation. On the real data we will report TTD_{ef} , FR, %Sup, and near-onset false suppression (FS@onset, Section 4.10) with the dataset’s own episode length and sample size, alongside the accident-anticipation metrics standard for that dataset. No real-data results are reported in this paper; this paragraph states the planned protocol only, and the entropy-lag tolerance measured in Section 4.10 sets the expectation that the scheduler should absorb the timing imprecision of real annotations up to a bounded lag.

6. Architectural Implications for Vehicular Edge Computing and Future Offloading

This section offers an architectural interpretation, not an evaluated distributed system. The campaign in Section 4 ran on a single embedded node under a fixed budget: no networked coordination, offloading, communication latency, or bandwidth analysis was evaluated. The validated contribution remains single-node embedded selective computation. Here we explain only how the node-level operational state signal ℓ_t , validated in that setting, could inform future VEC and MEC architectures.

6.1. TERA as a Source of Edge Operational State Signals

The MHEG runtime output is a discrete signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, emitted once per timestep at $O(1)$ cost (Algorithm 1, Eq. 13). It is

compact, a two-bit integer that can be stored, transmitted, or logged without reference to LSTM weights, recurrent state, or sensor frames, and it is interpretable through the epidemiological calibration of TERA-Gen (Table 3). When MHEG suppresses the update, ℓ_t comes from the buffered p_{t-1} with no inference at all. What sets ℓ_t apart from a thresholded classifier output is its temporal grounding. Under AF-TOI, ℓ_t reflects phase-conditioned trajectory structure rather than instantaneous confidence, and the negative control of Section 4.4 shows a Baseline scheduler without this structure degrades TTD_{ef} by 11.8%. This work’s contribution is producing ℓ_t with experimentally supported temporal grounding; how it is consumed externally is a downstream question for future work.

6.2. Candidate Integration Paths for VEC and MEC Offloading

Resource allocation at the embedded tier The most immediate implication is within the on-board unit. Under AF-TOI+MHEG, 48.7% of frames are suppressed without violating $\text{FR} \leq 0.05$, freeing cycles that could support other on-board functions. Whether such sharing is feasible depends on the hardware platform and RTOS scheduler, which this work does not evaluate.

Compatibility with VEC and MEC offloading In VEC and MEC architectures, the on-board unit decides which tasks to keep local and which to offload to roadside or cloud infrastructure. The signal ℓ_t could serve as one urgency input among others, with persistent Alert or Critical indicating time-critical conditions and Normal or Attention indicating deferrable ones. Such a policy could transmit only the compact symbol ℓ_t plus a summary feature window rather than raw frames, consistent with bandwidth-constrained partial offloading [53]. The activation fractions in Movement 6 (Section 4.7) are episode-level severity indicators, not measured communication rates; translating them into frame-level communication load requires network-level validation not performed here.

From scheduling signal to coordination primitive. In the borderline setting of Movement 6, activation fraction was more informative than instantaneous mean probability for summarizing the response to intermediate-risk episodes. A future policy could escalate on the persistence of elevated levels over a short window rather than on single readings, expressible as a rolling escalation statistic,

$$\Phi_t(w) = \frac{1}{w} \sum_{i=0}^{w-1} \mathbf{1}[\ell_{t-i} \geq 2], \quad (16)$$

the fraction of the last w timesteps at Alert or Critical. $\Phi_t(w)$ adds only a bounded moving sum over a symbol the system already emits, at a memory cost of w two-bit symbols, negligible next to the 1,024-byte buffer.

It is not evaluated here but offered as a candidate bridge between the validated single-node mechanism and a future multi-tier policy, consistent with distributed early-exit designs [54] and VEC offloading frameworks [53, 55]; a future system could combine it with channel quality, queue length, and latency budget. In summary, the single-node mechanism validated here is shaped, without modification, as a candidate coordination primitive for future VEC and MEC deployments, and the finding that activation frequency summarized escalating risk more stably than instantaneous mean probability could inform how a future escalation policy is designed. Empirical validation under real network conditions remains future work.

7. Conclusion

We set out to test whether selective computation can be made safe in strictly causal vehicular and ad hoc edge inference, and to identify what the safety actually depends on. Our hypothesis was that it depends less on the scheduler’s policy than on the temporal organization of the model’s predictive trajectories – the property that makes uncertainty informative in the seconds around a critical transition. Within the controlled setting studied here, the evidence supports that reading, and the negative control is what makes it credible: with the scheduler applied to unorganized trajectories, TTD_{ef} worsened by 11.8% and coverage did not improve, so activating a scheduler is not, by itself, enough. Given AF-TOI-organized trajectories, the same MHEG scheduler reached 53.4% effective cost reduction with TTD_{ef} of 0.101 s against 0.673 s. The post-hoc calibration control ruled out the competing explanation: probabilistic rescaling improved ECE without reorganizing entropy in time, so marginal calibration and temporally conditioned reliability are separate properties with separate origins. The design lesson is not specific to driver monitoring. A training protocol that optimizes only classification accuracy or aggregate calibration can still leave a model unsafe to schedule on, because neither objective constrains the internal trajectory structure a causal scheduler reads. On this evidence, temporal organization is worth treating as a design property in its own right for resource-constrained edge AI, rather than as an accidental byproduct of training. The pipeline ends in the operational-risk signal $\ell_t \in \{\text{Normal, Attention, Alert, Critical}\}$, produced at $O(1)$ cost by the chain *Temporal Organization* $\rightarrow$ *Entropy Stratification* $\rightarrow$ *MHEG*. Because it is compact, real-time, and interpretable at the level of the trajectory (Section 6.1), it is a plausible primitive for distributed and vehicular edge architectures to consume without touching the on-board loop – though we present that use as a design path (Section 6), not a validated result. The scope of the present validation should be stated explicitly. TERA is validated as a temporal-organization and scheduling substrate for a *single*

embedded node, under strictly causal inference, on controlled synthetic data with a deterministic onset; the whole deployed workflow fits in $\approx 58\text{K}$ parameters and 1,024 bytes of recurrent buffer, within reach of automotive-grade platforms. It is not validated as a standalone detection system, and no networked coordination, offloading, or multi-node behavior was evaluated. The VEC and MEC directions of Section 6 are implications of the ℓ_t signal, to be tested rather than claimed. Accordingly, the immediate priorities are external validity on naturalistic data, robustness under distributional shift, sensor faults, and adversarial perturbation, characterization on physical automotive hardware, and an empirical multi-node study of the offloading policies sketched here – each building on the single-node, entropy-gated mechanism this paper establishes.

Authors' Contribution Statement (CRediT)

Ângela Brígida Albarello: Conceptualization, Methodology, Software, Formal Analysis, Investigation, Writing (original draft, review and editing). **Edna Dias Canedo:** Academic Supervision, Conceptualization, Writing (review and editing). **Vinícius P. Gonçalves:** Writing (review and editing). **Leandro A. Villas:** Supervision, Writing (review and editing). **Rodolfo I. Meneguette:** Writing (review and editing). **Geraldo P. Rocha Filho:** General Supervision, Writing (review and editing).

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The authors declare no competing financial interests or personal relationships that could have influenced the reported results.

Data Availability

The source code, synthetic datasets, configuration files, and evaluation scripts will be made publicly available upon publication. For peer review, an anonymized repository will be made available to the reviewers upon request.

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